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(54) **ELECTRICAL CONNECTOR**

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(52) **U.S. Cl.**

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(57) **ABSTRACT**

A coaxial connector device includes: a signal contact member; a signal contact member; a ground contact member to which a ground potential is applied; and a conductive member including a rear ground connection portion that is connected to the ground contact member, and a front ground connection portion that is connected to a ground contact portion of a mating connector device. When viewed in a direction of fitting to the mating connector device, the front ground connection portion overlaps at least a part of a signal line including a center conductor of a coaxial cable and the signal contact member, and at least a part of a signal line including a center conductor of a coaxial cable and the signal contact member.

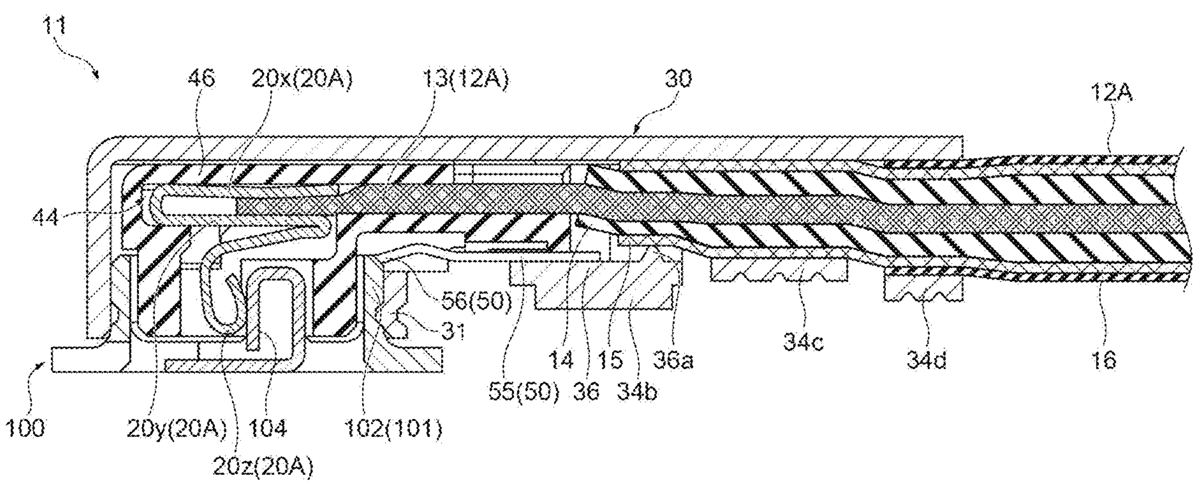
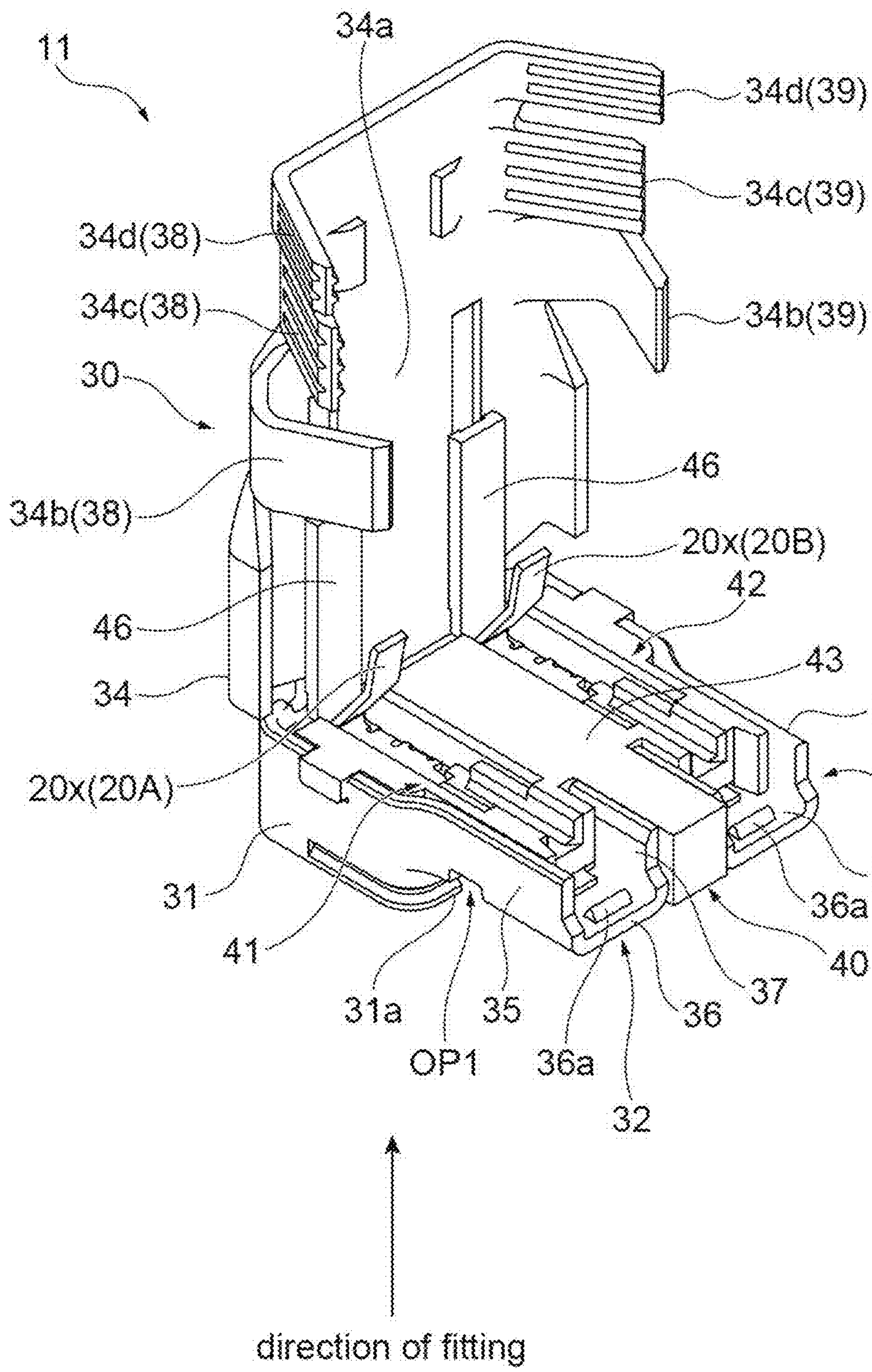


Fig.1



11
34a
34d(39)
34c(39)
34d(38)
34c(38)
34b(39)
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46
34b(38)
20x(20B)
42
46
43
34
35
33
20x(20A)
36
31
36a
41
40
37
31a
35
36
OP1
36a
32
direction of fitting

Fig.2

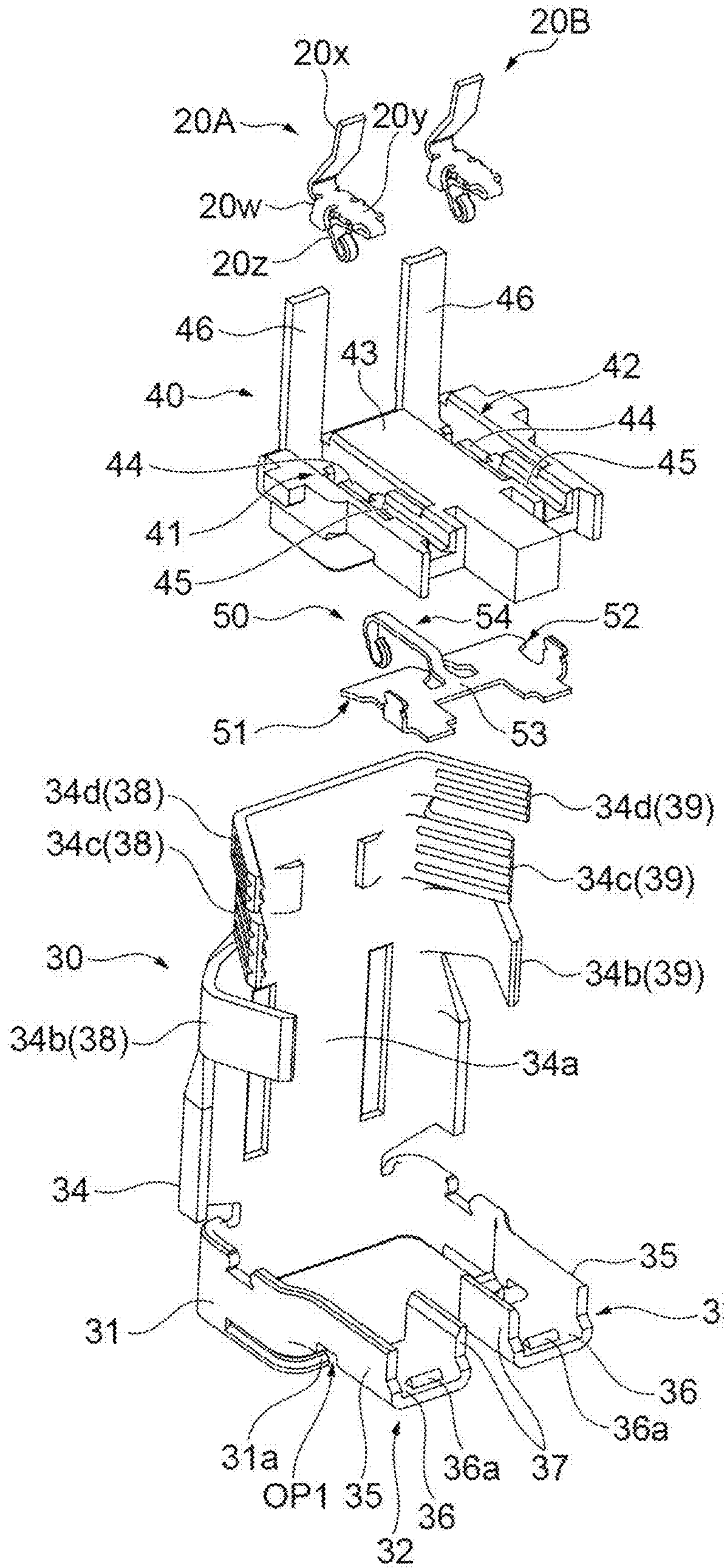
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Fig.3

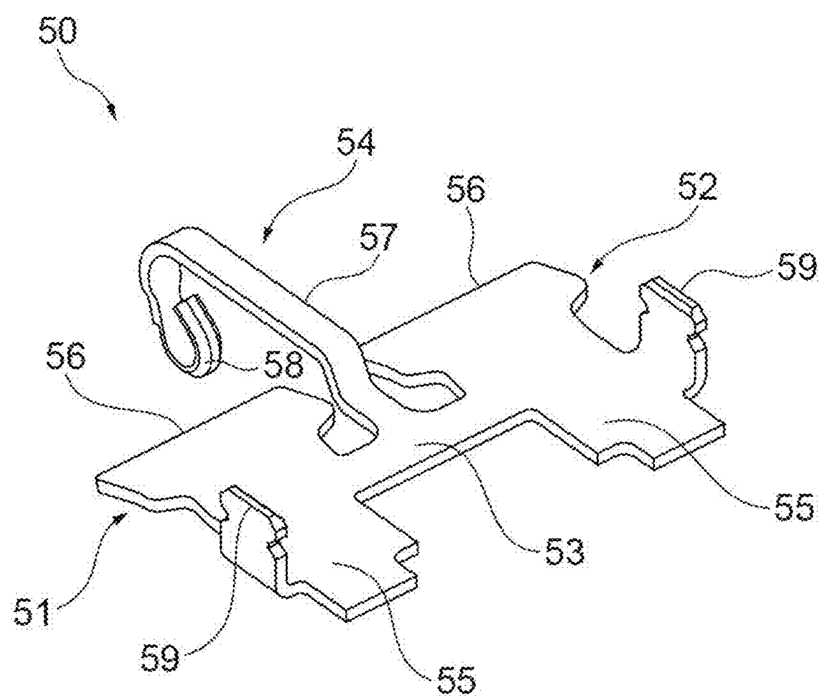
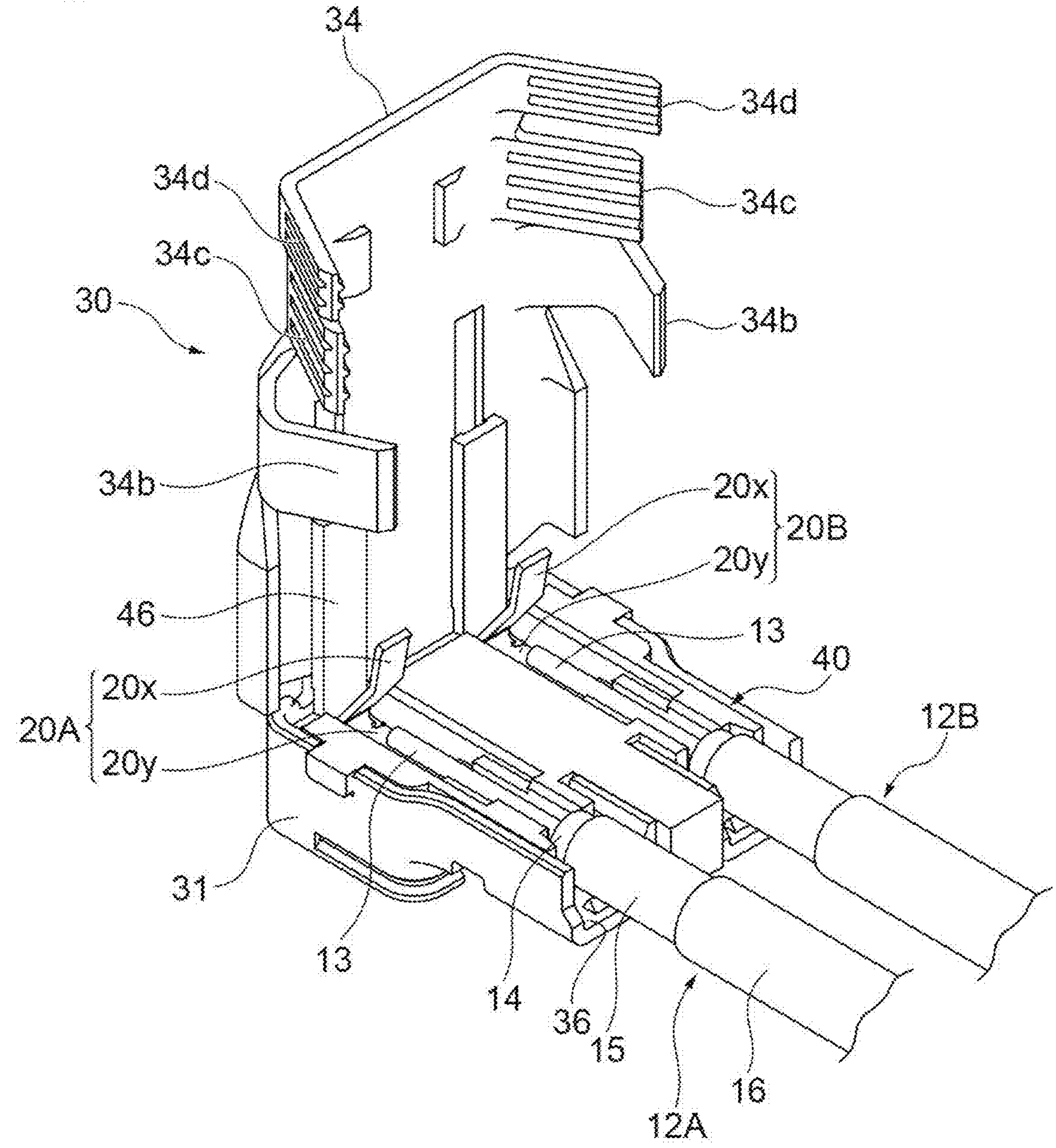


Fig.4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16
12A

Fig.5

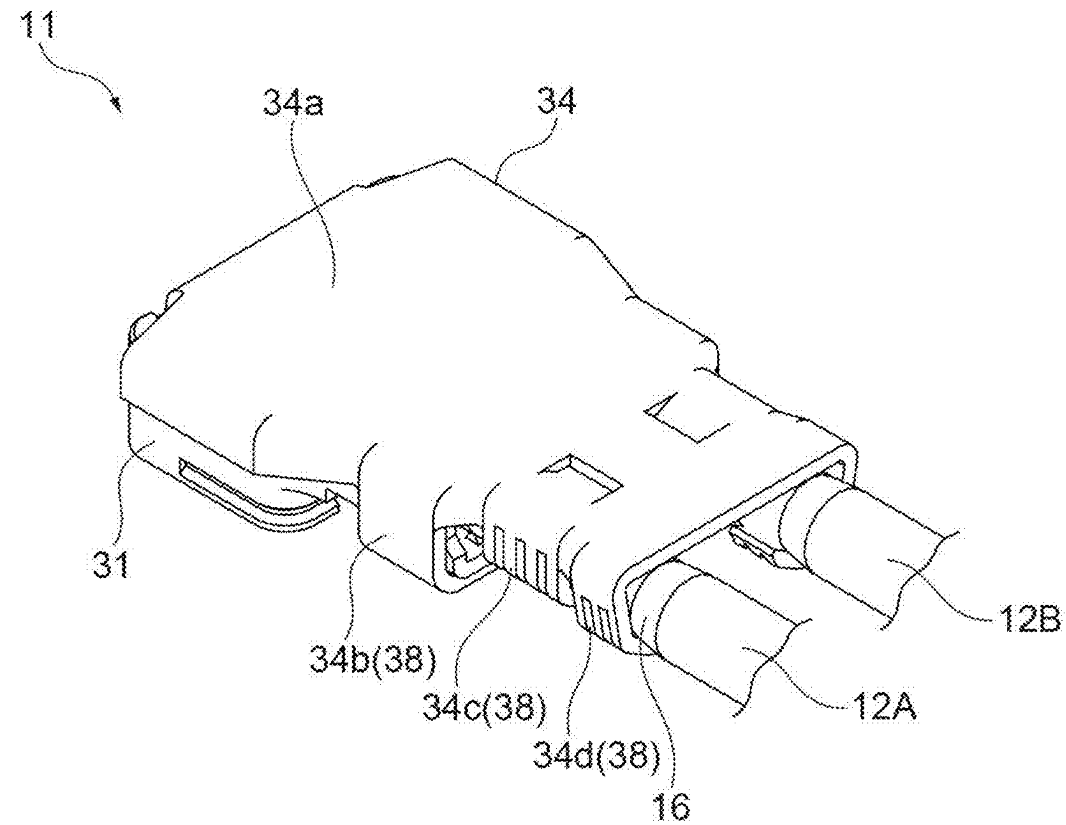
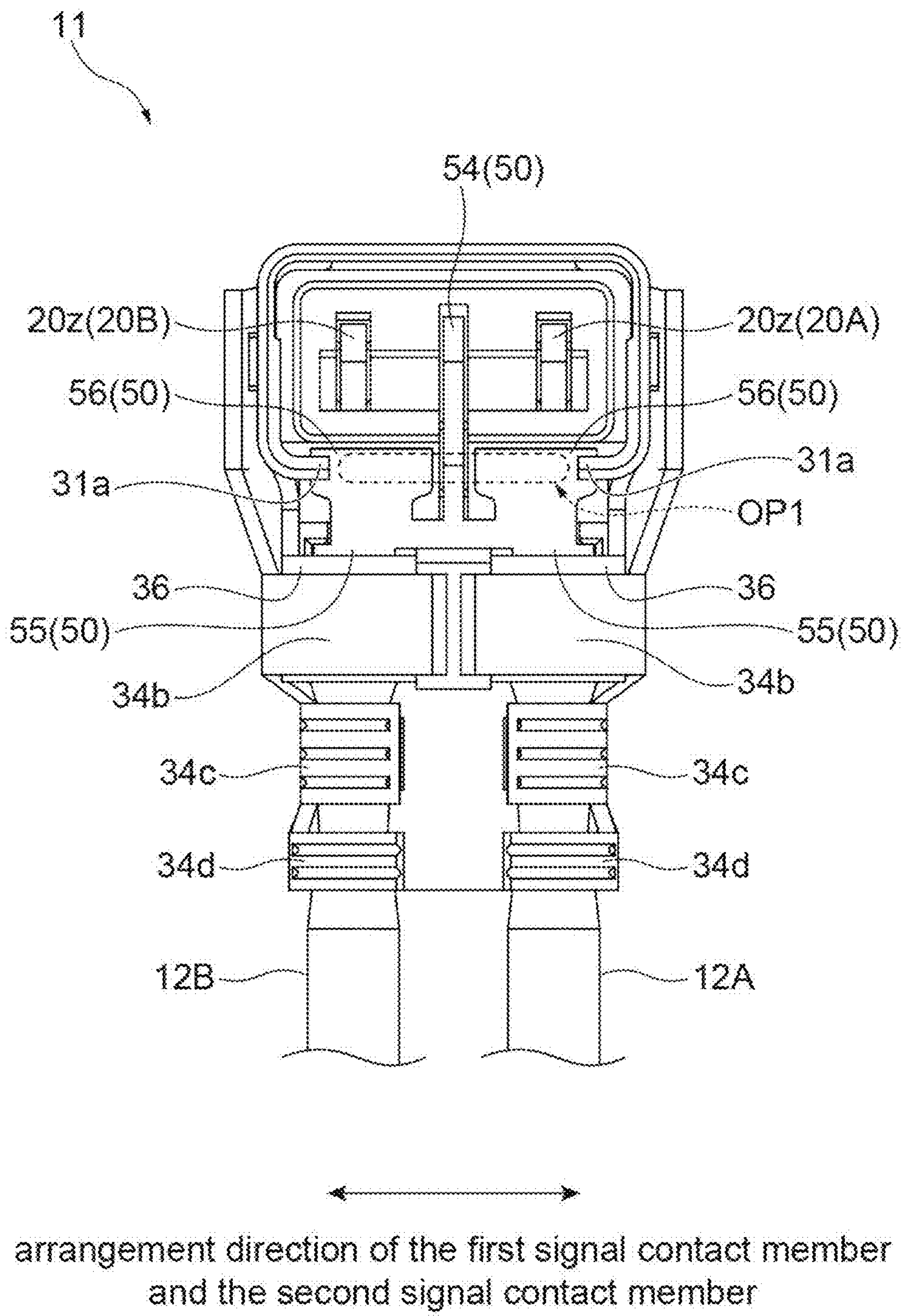


Fig.6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54(50)
20z(20B)
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31a
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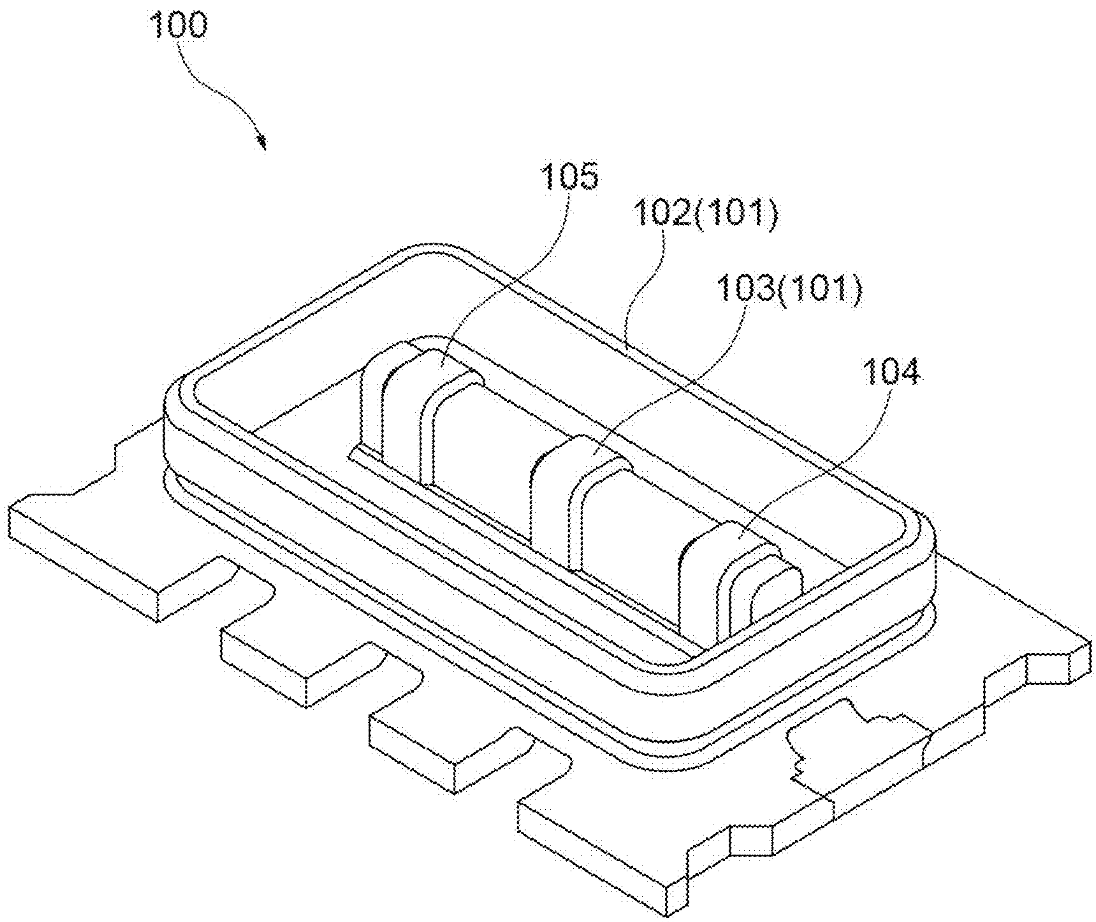
55(50)
55(50)
34b
34b
34c
34c
34d
34d
12B
12A
arrangement direction of the first signal contact member
and the second signal contact member

Fig.7



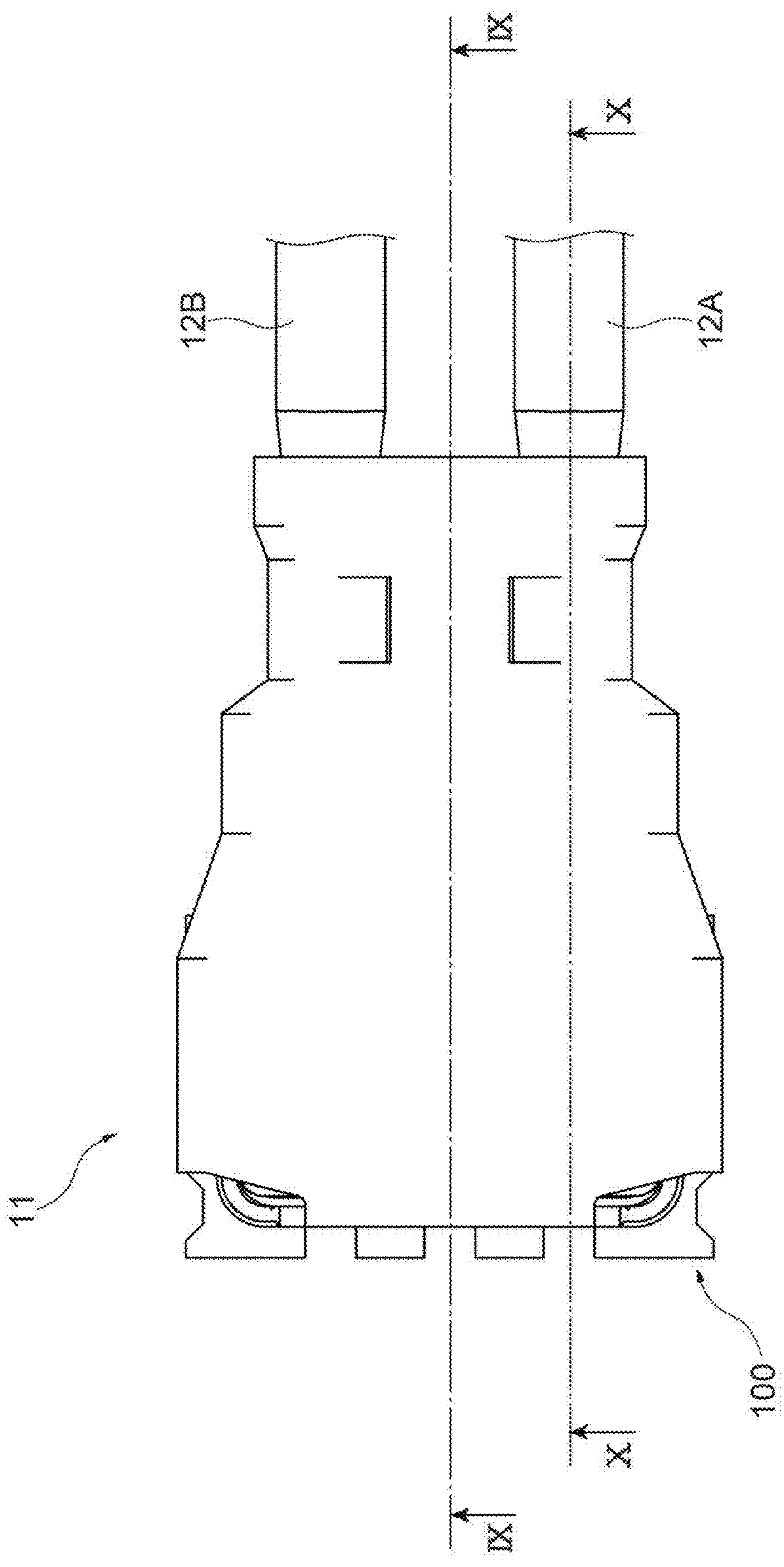


Fig.8

Fig.9

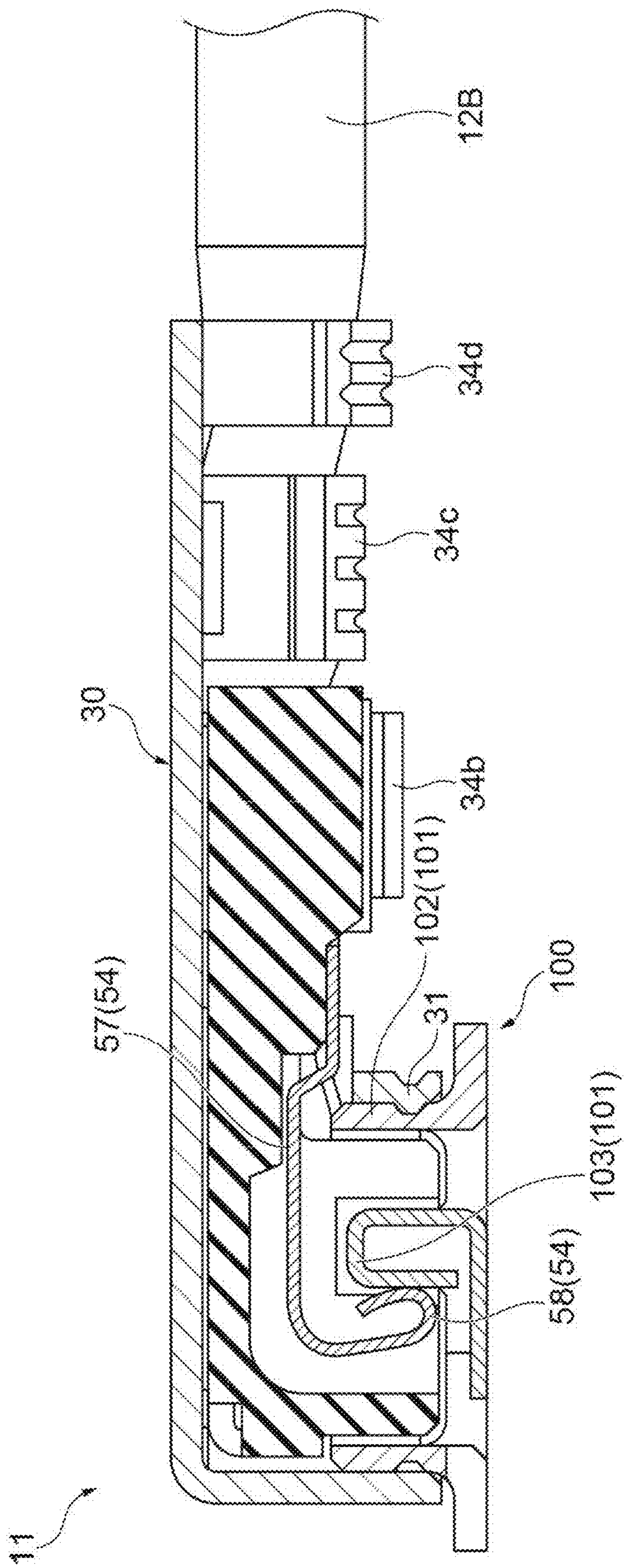
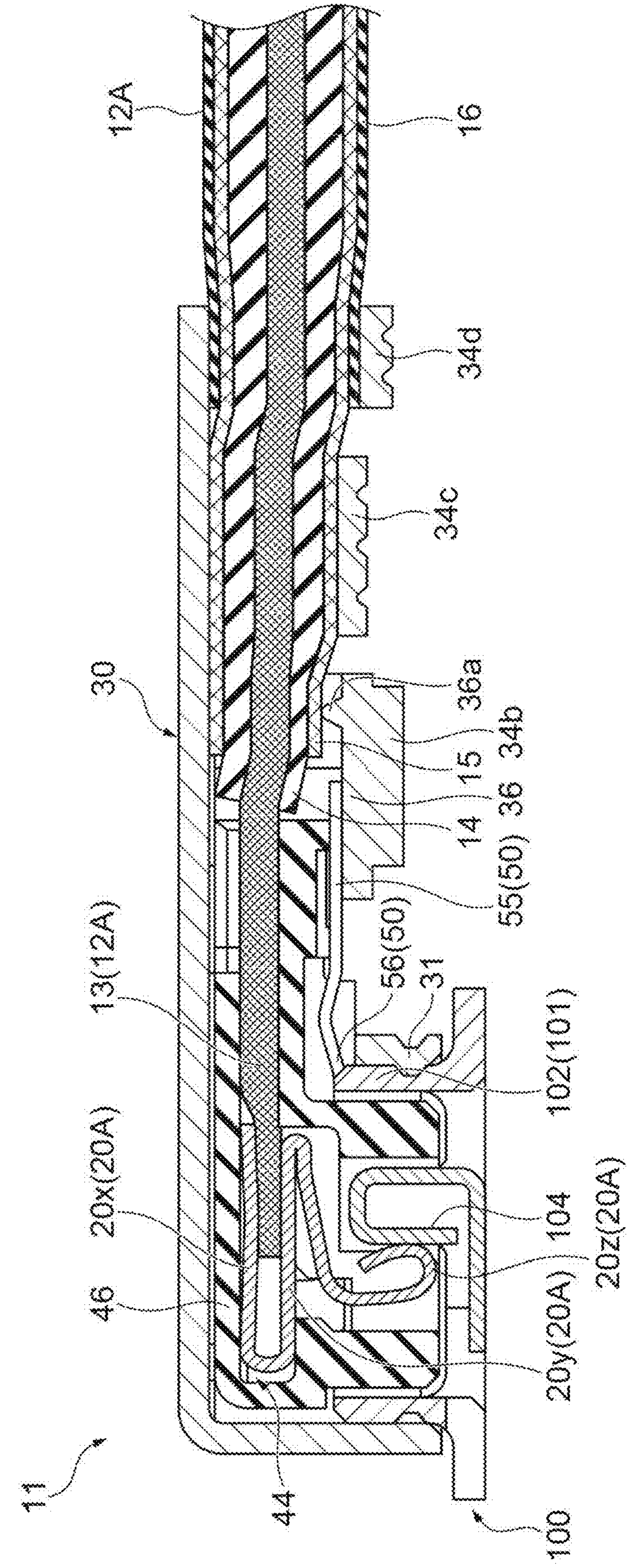


Fig. 10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34d
34c
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36a
34b
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13(12A)
56(50)
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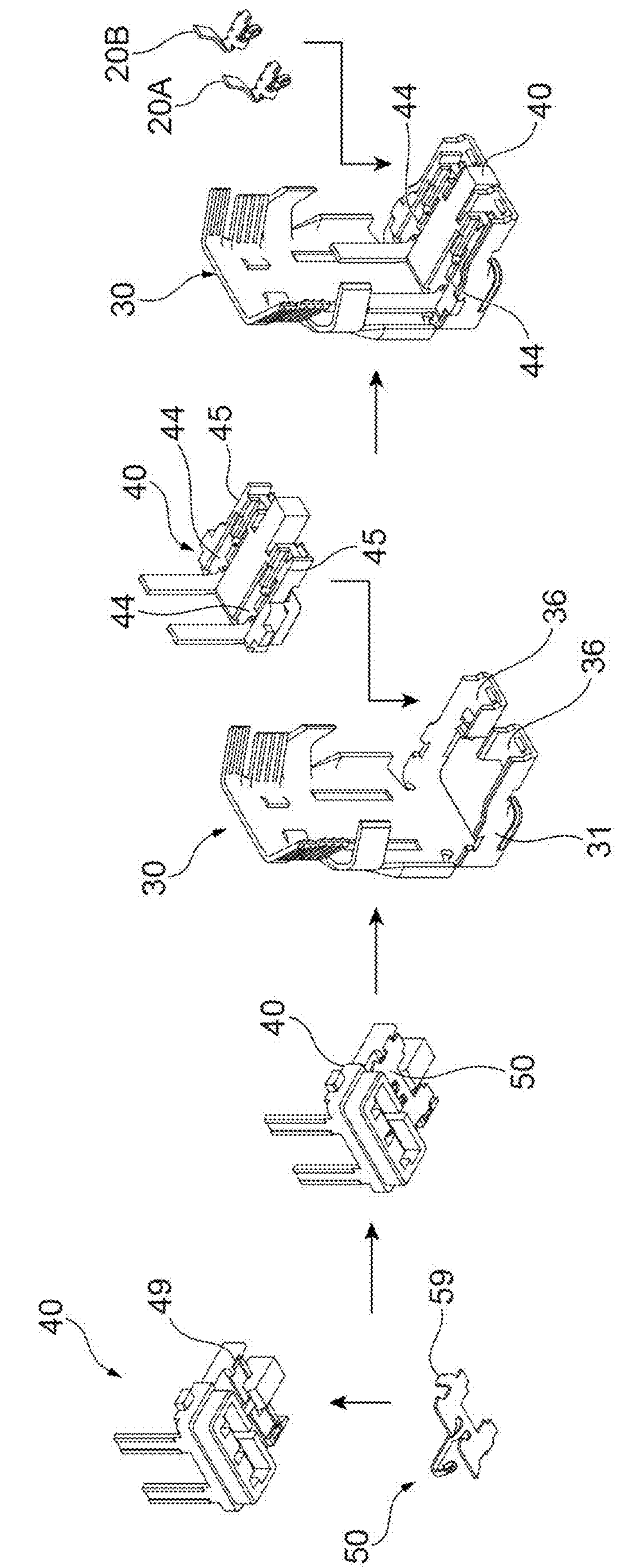
104
20z(20A)
46
20y(20A)
44
11
100

Fig.11A Fig.11B Fig.11C Fig.11D



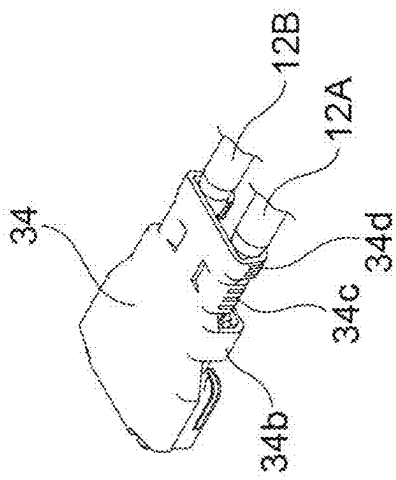


Fig.12C

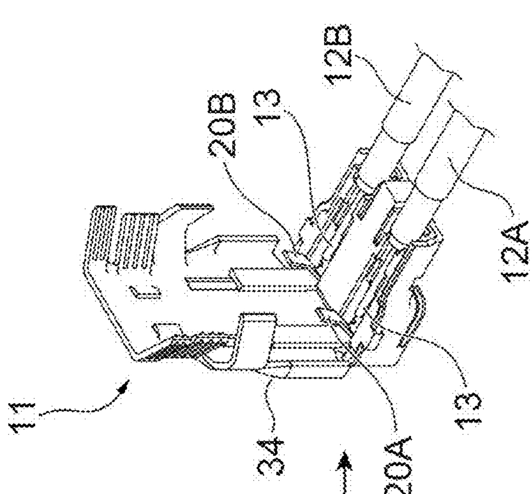


Fig.12B

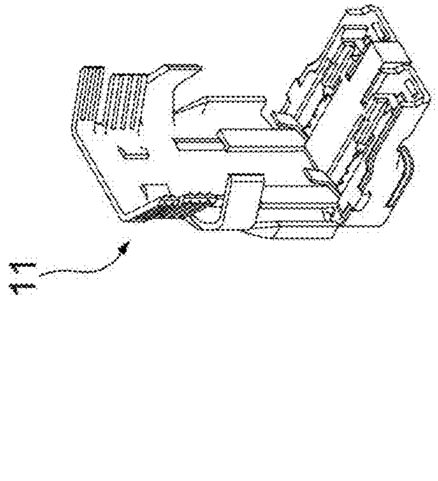
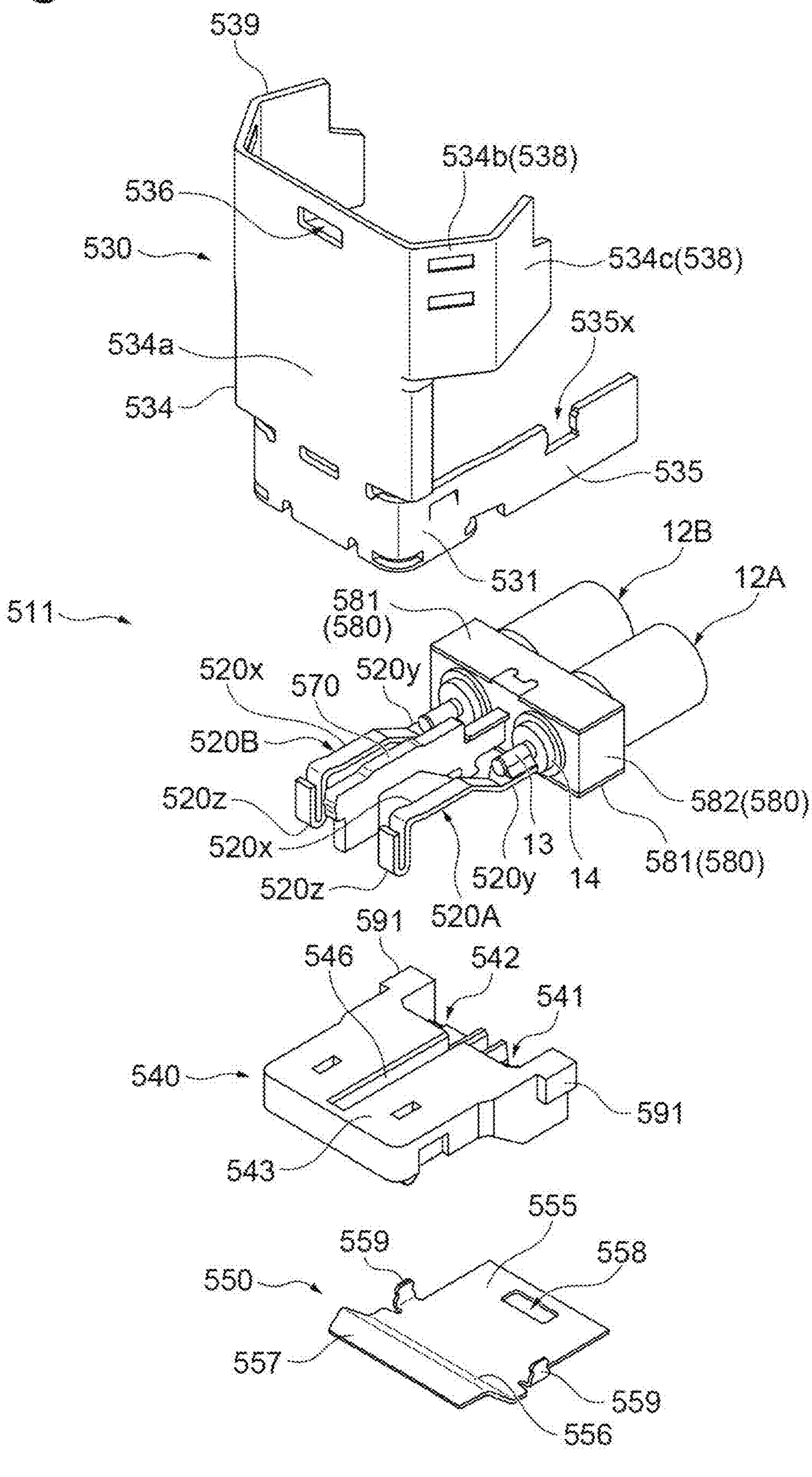


Fig.12A

Fig.13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(580)
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520B
520z
582(580)
520x
13
520y
14
581(580)
520z
520A
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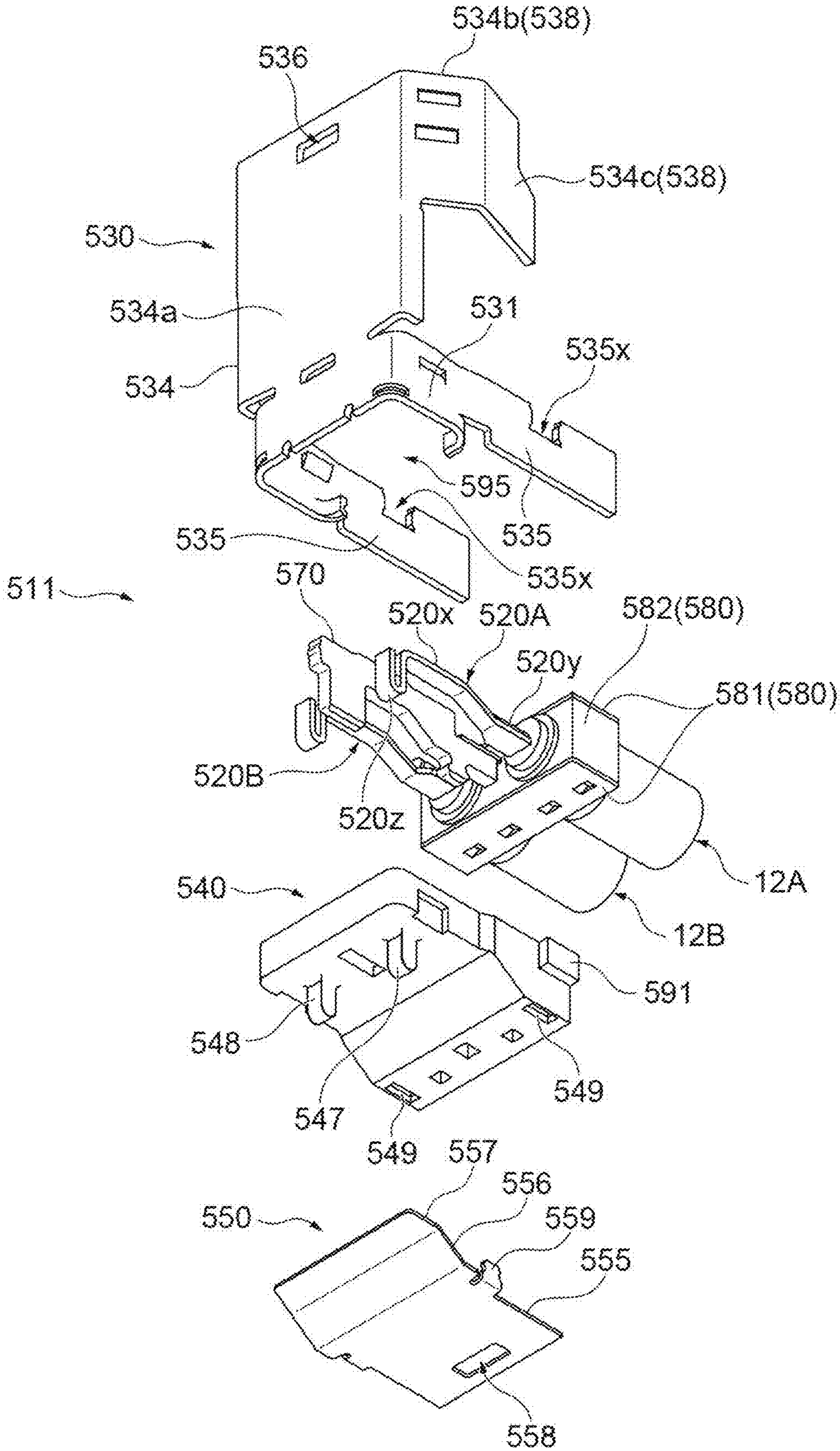
Fig.14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535x
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520x
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520y
581(580)
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Fig.15A

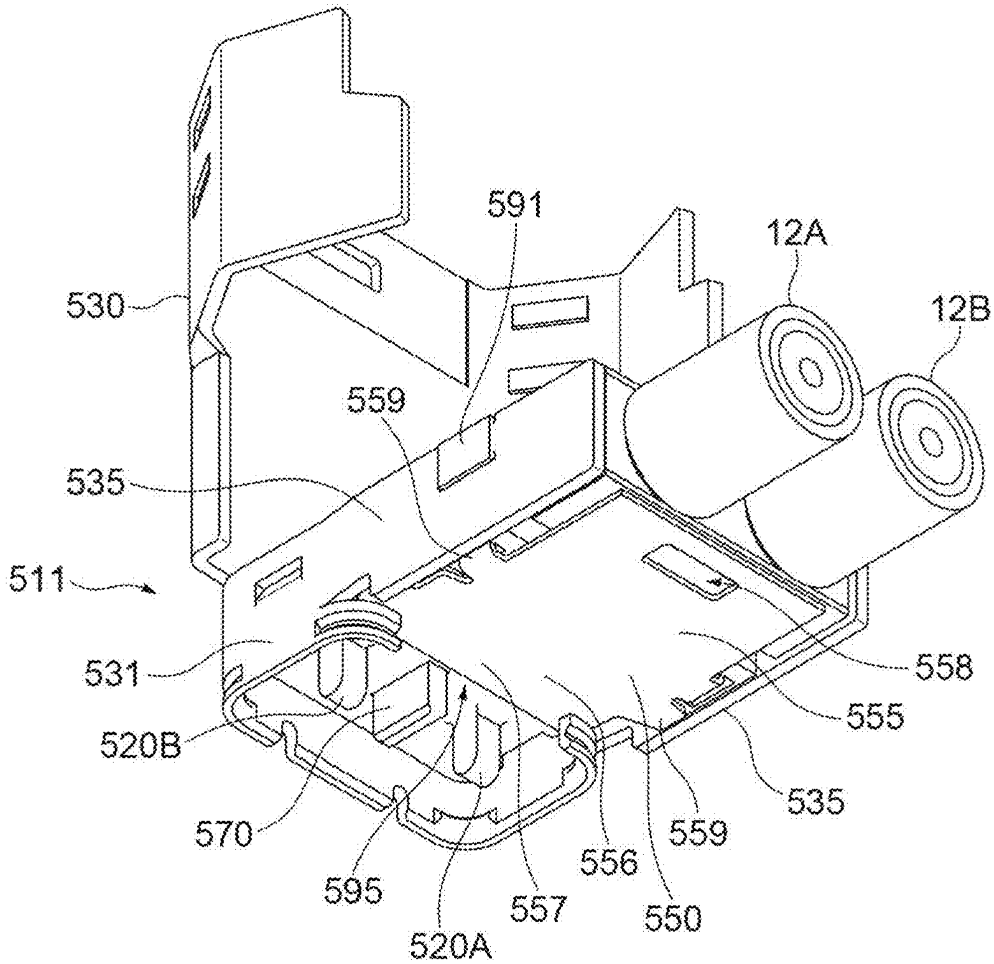


Fig.15B

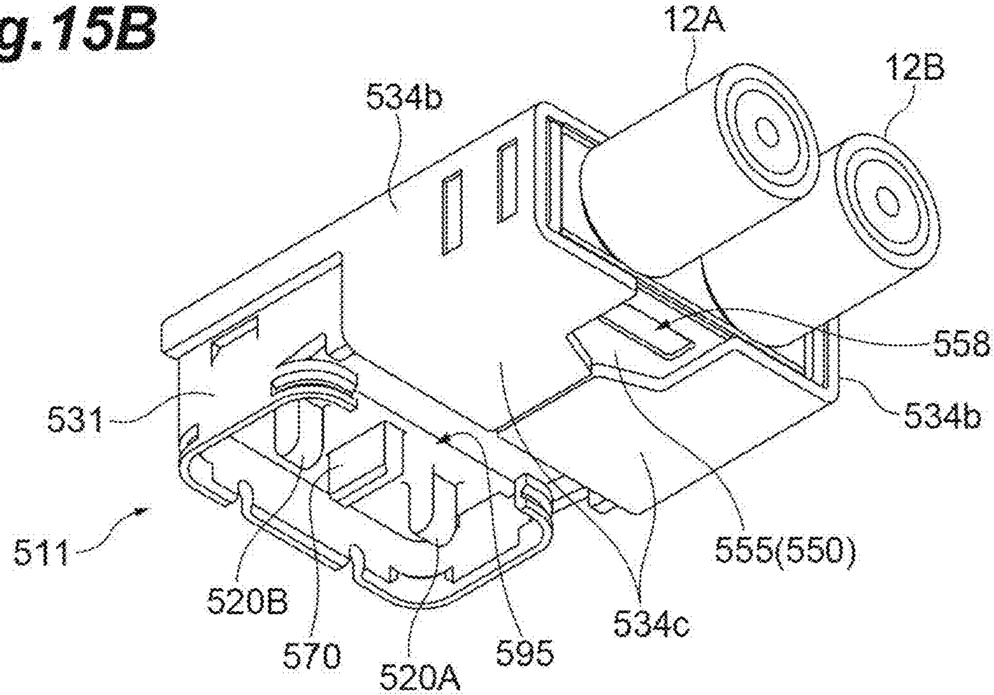


Fig.16

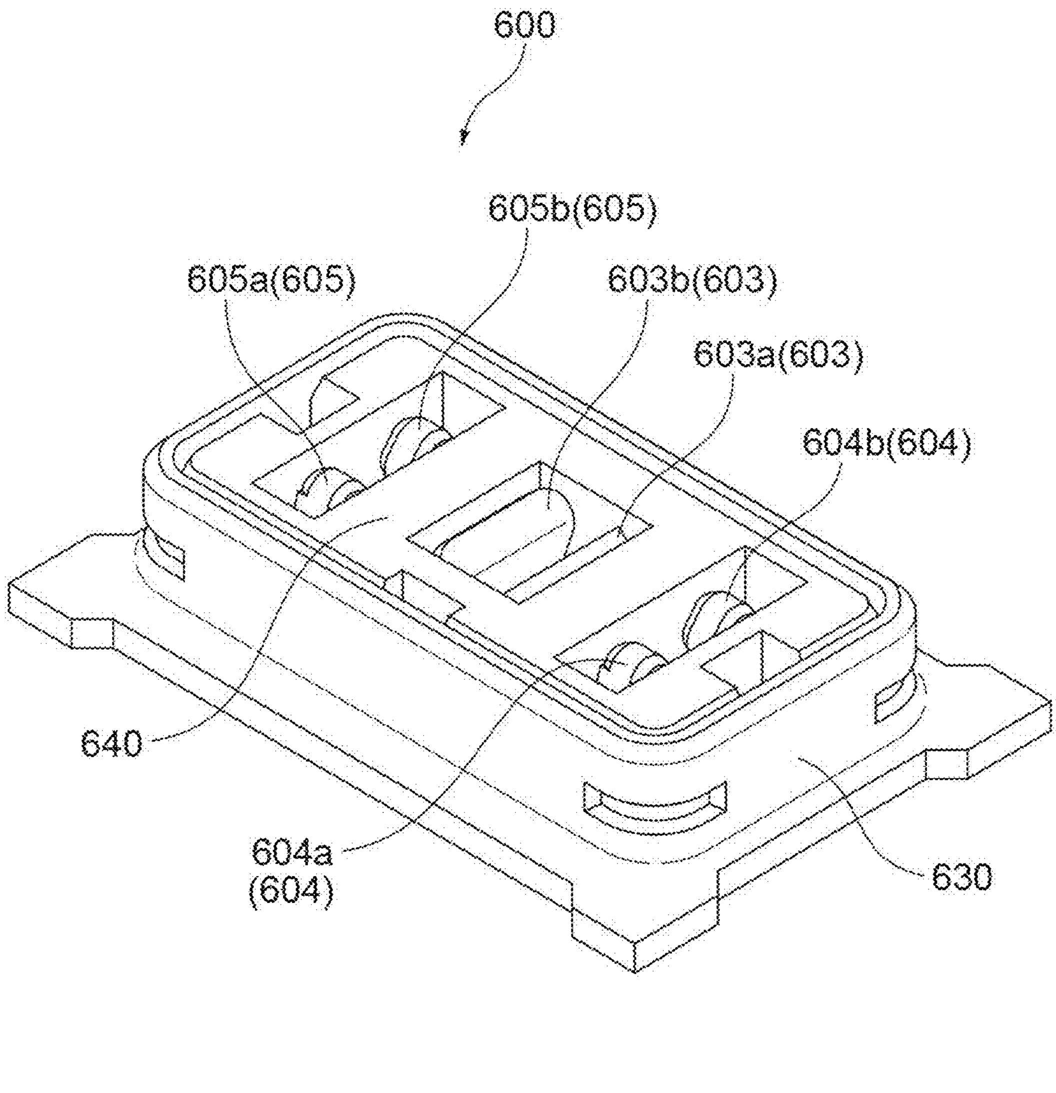

600
605b(605)
605a(605)
603b(603)
603a(603)
604b(604)
640
604a
(604)
630

Fig.17

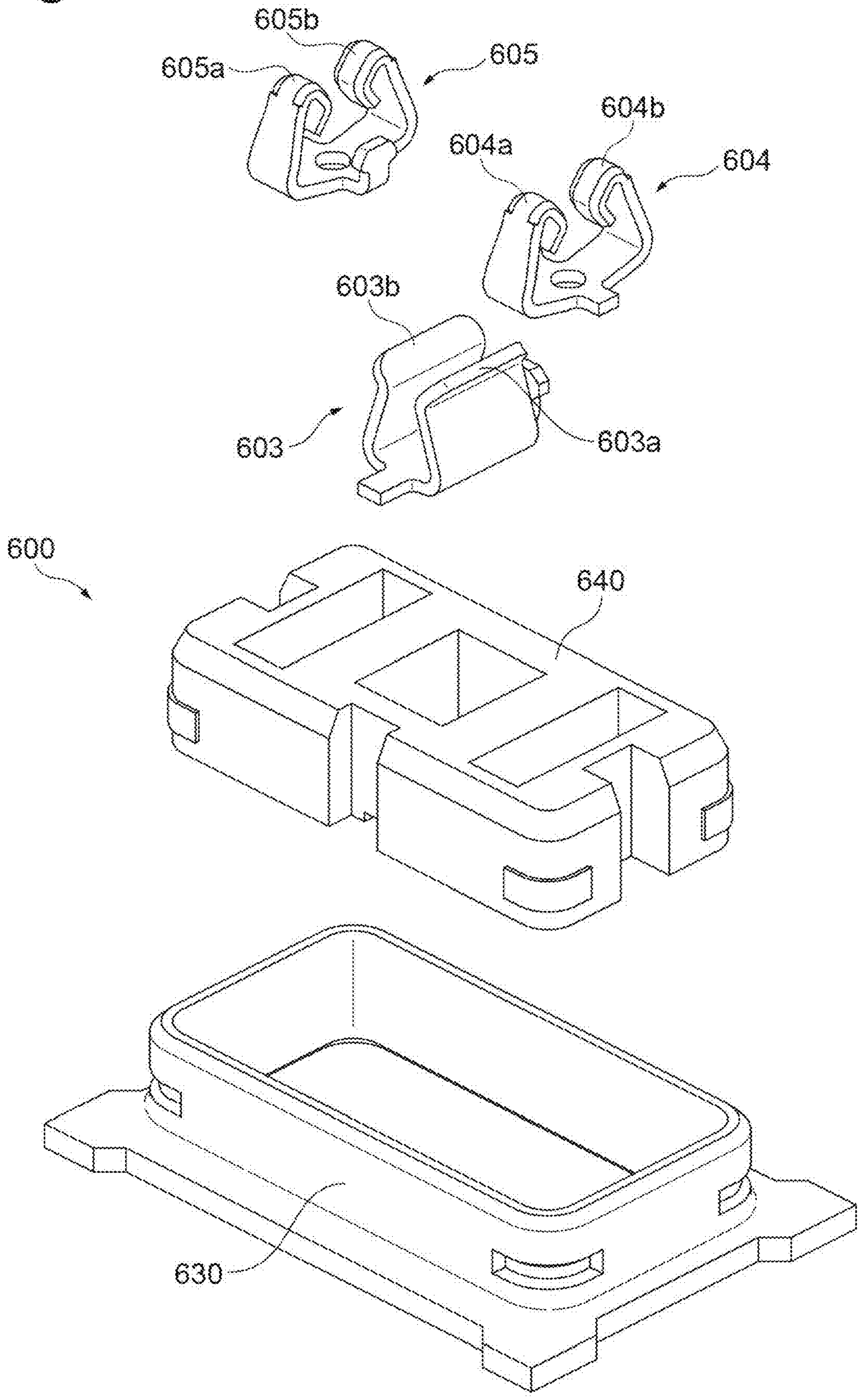


Fig.18A

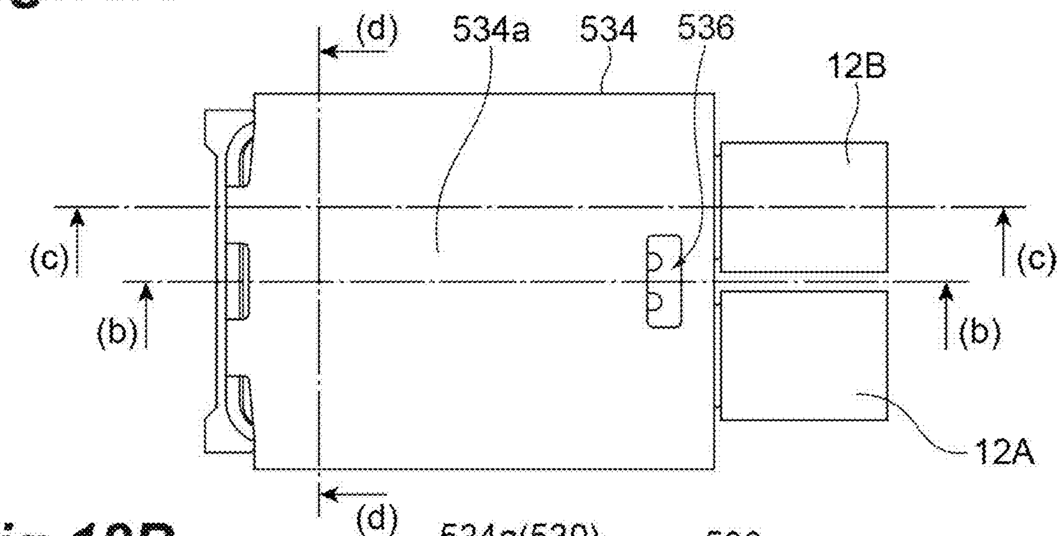


Fig.18B

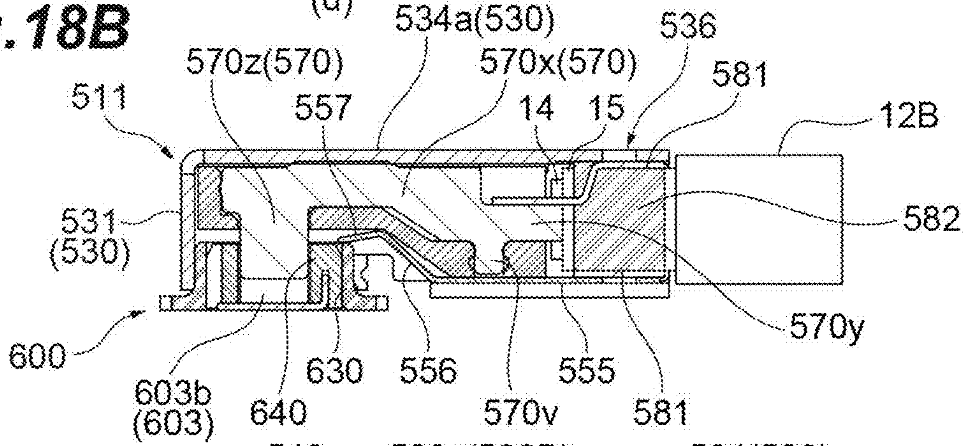


Fig.18C

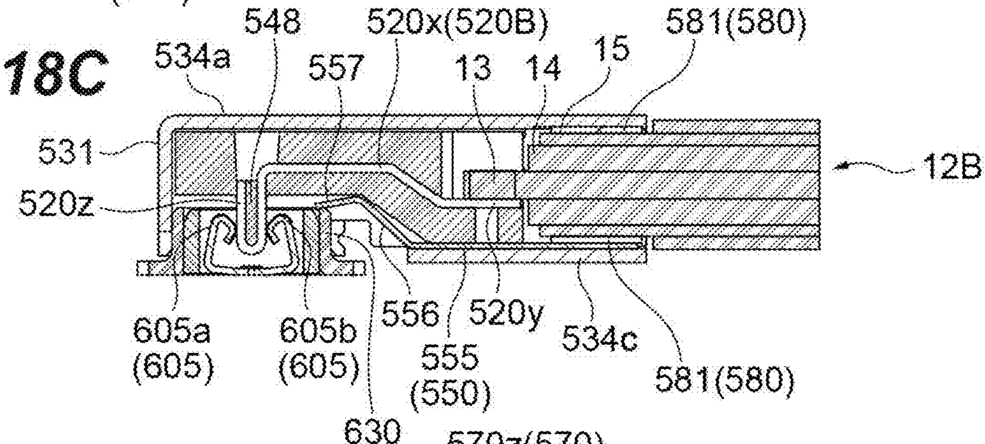


Fig.18D

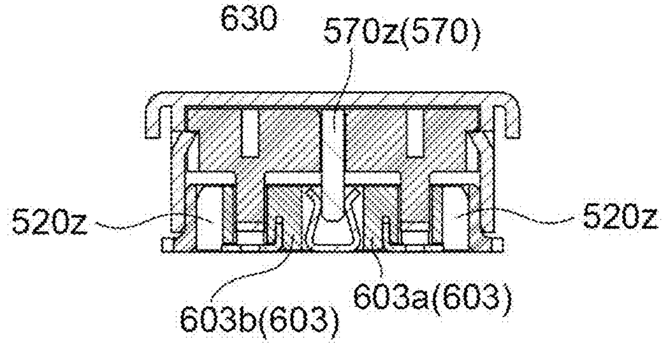


Fig.19A Fig.19B Fig.19C Fig.19D

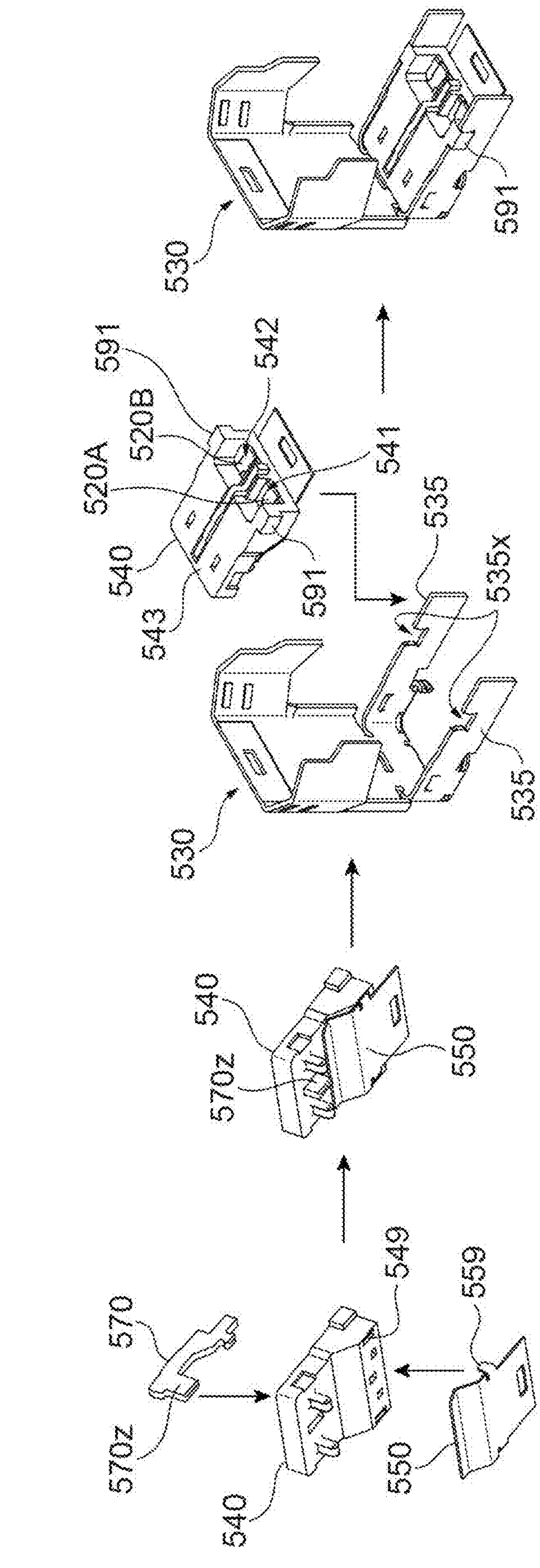


Fig.20A

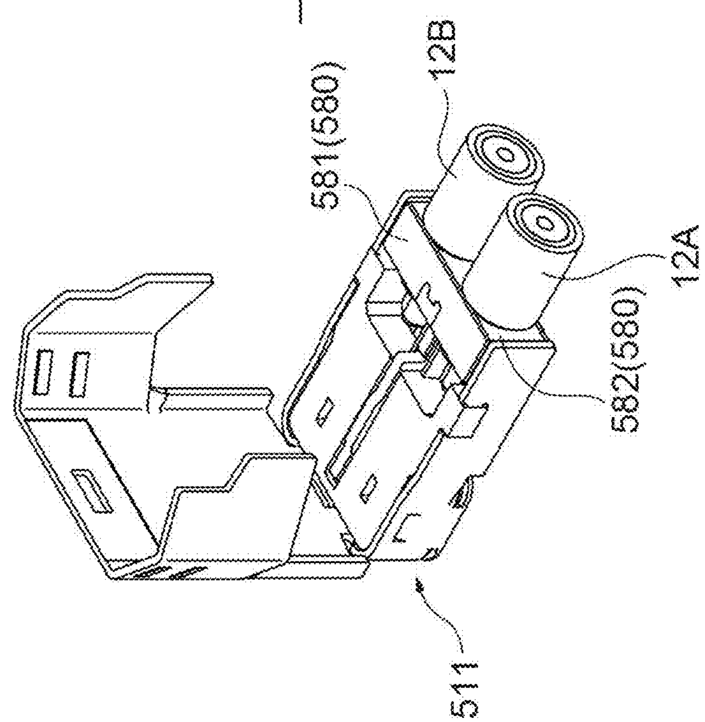
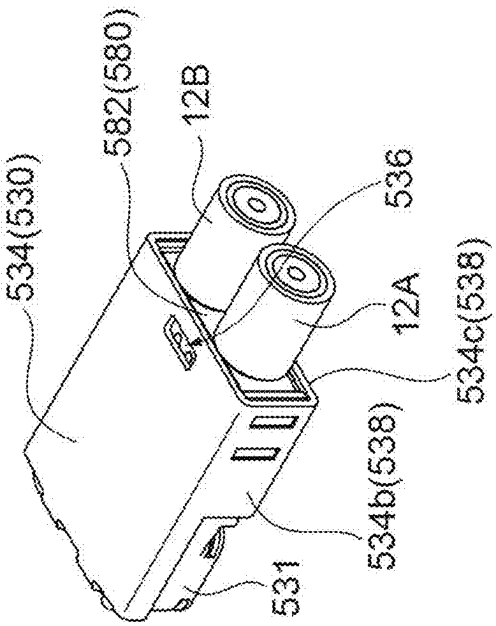


Fig.20B



ELECTRICAL CONNECTOR

TECHNICAL FIELD

[0001] The present invention relates to an electrical connector.

BACKGROUND ART

[0002] A connector device has been known in which a plug connector attached to a terminal portion of a coaxial cable is fitted to a receptacle connector mounted on a wiring board to electrically connect the coaxial cable and an electrical circuit of the wiring board (refer to, for example, Patent Literature 1).

CITATION LIST

Patent Literature

[0003] Patent Literature 1: Japanese Patent No. 6269558

SUMMARY OF INVENTION

Technical Problem

[0004] Here, in the connector device as described above, when signals inside the connectors propagate to surroundings of the connector device, the signals affect surrounding external products, which is a concern. For this reason, in the connector device as described above, it is important to prevent signals inside the connectors from propagating to the surroundings (to improve noise resistance).

[0005] The present invention is conceived in view of the above circumstances, and an object of the present invention is to provide an electrical connector capable of improving noise resistance.

Solution to Problem

[0006] An electrical connector according to one aspect of the present invention includes: a first signal contact member that is connected to a center conductor of a first coaxial cable, and that is connected to a first signal contact portion of a mating connector mounted on a circuit board; a second signal contact member that is connected to a center conductor of a second coaxial cable, and that is connected to a second signal contact portion of the mating connector; a ground contact member which includes an annular fitting portion that is fitted and connected to a ground contact portion of the mating connector, and a shell portion that bendably extends from the annular fitting portion to be connected to outer conductors of the first coaxial cable and of the second coaxial cable, and to which a ground potential is applied; and a conductive member including a first ground connection portion that is connected to the ground contact member, and a second ground connection portion that is connected to the ground contact portion of the mating connector. When viewed in a direction of fitting to the mating connector, the second ground connection portion of the conductive member overlaps at least a part of a first signal line including the center conductor of the first coaxial cable and the first signal contact member, and at least a part of a second signal line including the center conductor of the second coaxial cable and the second signal contact member.

[0007] The electrical connector according to one aspect of the present invention includes the first signal contact member connected to the center conductor of the first coaxial cable; the second signal contact member connected to the center conductor of the second coaxial cable; the ground contact member to which a ground potential is applied; and the conductive member. The first ground connection portion of the conductive member is connected to the ground contact member, and the second ground connection portion is connected to the ground contact portion of the mating connector. Furthermore, in the electrical connector, when viewed in the direction of fitting, the second ground connection portion overlaps at least a part of the first signal line including the center conductor of the first coaxial cable and the first signal contact member, and at least a part of the second signal line including the center conductor of the second coaxial cable and the second signal contact member. Since the second ground connection portion overlaps at least a part of the first and second signal lines, signals of the first and second signal lines are prevented from propagating to the outside (surroundings of a connector device including the electrical connector and the mating connector). As a result, signals of the first and second signal lines can be prevented from becoming noise and affecting surrounding external products, EMI characteristics of the electrical connector can be improved, and noise resistance can be improved.

[0008] The shell portion of the ground contact member may include barrel portions that clamp and hold the first coaxial cable and the second coaxial cable, respectively, and the conductive member may be connected to the barrel portions. In such a manner, since the barrel portions that hold the coaxial cables forming the signal lines and the conductive member are connected to (in contact with) each other, contact locations can be reliably set to a ground potential, and portions at which the signal lines and the conductive member (specifically, the second ground connection portion) overlap each other can be suitably formed.

[0009] The ground contact portion of the mating connector may include a first portion that is fitted and connected to the annular fitting portion of the ground contact member. The second ground connection portion may include a pair of first connection portions that are connected to the first portion. One of the pair of first connection portions may overlap at least a part of the first signal line when viewed in the direction of fitting, and the other of the pair of first connection portions may overlap at least a part of the second signal line when viewed in the direction of fitting. In such a manner, since the pair of first connection portions of the second ground connection portion are connected to a portion related to fitting (first portion) in the ground contact portion of the mating connector, in a state where the pair of first connection portions are fitted to the mating connector, a contact location can be reliably set to a ground potential. Furthermore, since one of the pair of first connection portions overlaps at least a part of the first signal line, and the other of the pair of first connection portions overlaps at least a part of the second signal line, signals of the first and second signal lines can be reliably prevented from propagating to the outside (surroundings of the connector device including the electrical connector and the mating connector).

[0010] The shell portion of the ground contact member may include barrel portions that clamp and hold the first coaxial cable and the second coaxial cable, respectively. When viewed in the direction of fitting, the pair of first connection portions may overlap the barrel portions in an arrangement direction of the first signal contact member and the second signal contact member. In such a manner, since the pair of first connection portions and the barrel portions that hold the coaxial cables overlap each other, portions at which the pair of first connection portions and the signal lines (specifically, the center conductors of the coaxial cables) overlap each other can be suitably formed. As a result, signals of the first and second signal lines can be appropriately prevented from propagating to the outside (surroundings of the connector device including the electrical connector and the mating connector).

[0011] An opening portion may be formed in the annular fitting portion of the ground contact member, and the pair of first connection portions may be disposed in the opening portion. In such a manner, for example, since the pair of first connection portions forming the conductive member are provided in the opening portion that is formed in a process of manufacturing, the opening portion is partially closed by the pair of first connection portions, so that signals (high-frequency signals) of the signal lines can be effectively prevented from leaking from the opening portion to the outside.

[0012] The ground contact portion of the mating connector may include a second portion disposed between the first signal contact portion and the second signal contact portion, and the second ground connection portion may include a second connection portion that is connected to the second portion. Since the second connection portion is connected to the second portion of the ground contact portion of the mating connector disposed between the first signal contact portion and the second signal contact portion, signals are prevented from propagating between the signal contacts that are different from each other. As a result, an isolation characteristic can be improved, and noise resistance can be improved.

[0013] The shell portion of the ground contact member may include barrel portions that hold the first coaxial cable and the second coaxial cable, respectively, and the conductive member may be connected to the barrel portions. In such a manner, since the barrel portions that hold the coaxial cables forming the signal lines and the conductive member are connected to (in contact with) each other, contact locations can be reliably set to a ground potential, and portions at which the signal lines and the conductive member (specifically, the second ground connection portion) overlap each other can be suitably formed.

[0014] The ground contact portion of the mating connector may include a first portion that is fitted and connected to the annular fitting portion of the ground contact member. The second ground connection portion may include a first connection portion that is connected to the first portion. The first connection portion may be continuously formed in an arrangement direction of the first signal contact member and the second signal contact member, and overlap at least a part of the first signal line and at least a part of the second signal line when viewed in the direction of fitting. In such a manner, since the first connection portion of the second ground connection portion is connected to a portion related to fitting (first portion) in the ground contact portion of the mating connector, in a state where the first connection portion is fitted to the mating connector, a contact location can be reliably set to a ground potential. Furthermore, since the first connection portion that is continuously formed without a cutout, a gap, or the like formed overlaps at least a part of the first signal line and at least a part of the second signal line, signals of the first and second signal lines can be more appropriately prevented from propagating to the outside (surroundings of the connector device including the electrical connector and the mating connector).

[0015] An opening portion may be formed in the annular fitting portion of the ground contact member, and the first connection portion may be disposed in the opening portion. In such a manner, for example, since the first connection portion forming the conductive member is provided in the opening portion that is formed in a process of manufacturing, the opening portion is effectively closed by the first connection portion, so that signals (high-frequency signals) of the signal lines can be effectively prevented from leaking from the opening portion to the outside.

[0016] The second connection portion may be disposed between the first signal contact member and the second signal contact member to block at least a part of a region between the first signal contact member and the second signal contact member. In such a manner, since the second connection portion is disposed between the signal contact members to block at least a part of the region between the signal contact members, signals can be more effectively prevented from propagating between the signal contact members. Accordingly, the isolation characteristic can be further improved.

[0017] The second connection portion may extend along an extending direction of the first signal contact member and of the second signal contact member. In such a manner, since the second connection portion extends along the extending direction of the first signal contact member and of the second signal contact member, the region between the signal contact members is effectively blocked by the second connection portion, so that the above-described isolation characteristic can be further improved.

[0018] The second connection portion may be disposed between the first signal contact member and the second signal contact member to block an entirety of the region between the first signal contact member and the second signal contact member. In such a manner, since the region between the signal contact members is completely blocked by the second connection portion, the above-described isolation characteristic can be further improved.

[0019] An area of the second connection portion when viewed in an arrangement direction of the first signal contact member and the second signal contact member may be larger than an area of each of the first signal contact member and the second signal contact member when viewed in the arrangement direction. In such a manner, since the area of the second connection portion between the signal contact members is set to be large, signals are more effectively prevented from propagating between the signal contact members by the second connection portion. Accordingly, the isolation characteristic can be further improved.

[0020] The second connection portion may be formed of a member separate from other regions in the second ground connection portion. Accordingly, the degree of freedom in the shape, the raw material, and the disposition of the second connection portion is increased, and signals are more effectively prevented from propagating between the signal contact members by the second connection portion of which the shape, the raw material, and the disposition are appropriately set. Accordingly, the isolation characteristic can be further improved.

[0021] The second connection portion may be connected to a ground bar electrically connected to the outer conductors of the first coaxial cable and of the second coaxial cable. Accordingly, similarly to the center conductor of the first coaxial cable and the center conductor of the second coaxial cable adjacent to each other with the outer conductors sandwiched therebetween, the first signal contact member and the second signal contact member are adjacent to each other with the second connection portion sandwiched therebetween, the second connection portion being electrically connected to the outer conductors through the ground bar, so that similarly to the coaxial cables adjacent to each other, signals can be effectively prevented from propagating between the signal contact members.

Advantageous Effects of Invention

[0022] According to one aspect of the present invention, noise resistance can be improved.

BRIEF DESCRIPTION OF DRAWINGS

[0023] FIG. **1** is a perspective view showing a coaxial connector device according to the present embodiment.

[0024] FIG. **2** is an exploded perspective view showing the coaxial connector device according to the present embodiment.

[0025] FIG. **3** is a perspective view of a conductive member of the coaxial connector device according to the present embodiment when viewed from above.

[0026] FIG. **4** is a perspective view showing a process of attaching the coaxial connector device according to the present embodiment to one end portions of coaxial cables.

[0027] FIG. **5** is a perspective view from above showing a state where the coaxial connector device according to the present embodiment is attached to the one end portions of the coaxial cables.

[0028] FIG. **6** is a bottom view showing a state where the coaxial connector device according to the present embodiment is attached to the one end portions of the coaxial cables.

[0029] FIG. **7** is a perspective view showing a mating connector device to which the coaxial connector device according to the present embodiment is coupled.

[0030] FIG. **8** is a plan view showing a state where the coaxial connector device according to the present embodiment is attached to the one end portions of the coaxial cables and is mechanically and electrically coupled to the mating connector device.

[0031] FIG. **9** is a cross-sectional view showing a cross section taken along line IX-IX in FIG. **8**.

[0032] FIG. **10** is a cross-sectional view showing a cross section taken along line X-X in FIG. **8**.

[0033] FIGS. **11**A to **11**D are views for describing a method for assembling the coaxial connector device according to the present embodiment.

[0034] FIGS. **12**A to **12**C are views for describing the method for assembling the coaxial connector device according to the present embodiment, and are views showing steps following FIG. **11**D.

[0035] FIG. **13** is an exploded perspective view from above showing a coaxial connector device according to another aspect of the present embodiment.

[0036] FIG. **14** is an exploded perspective view from below showing the coaxial connector device of FIG. **13**.

[0037] FIG. **15**A is a perspective view from below showing a state where the coaxial connector device of FIG. **13** is attached to one end portions of coaxial cables, and is a view showing a state before a shell portion holds a cable fixing portion, and FIG. **15**B is a perspective view from below showing a state where the coaxial connector device of FIG. **13** is attached to the one end portions of the coaxial cables, and is a view showing a state where the shell portion holds the cable fixing portion.

[0038] FIG. **16** is a perspective view showing a mating connector device to which the coaxial connector device of FIG. **13** is coupled.

[0039] FIG. **17** is an exploded perspective view from above showing the mating connector device of FIG. **16**.

[0040] FIG. **18**A is a plan view showing a state where the coaxial connector device of FIG. **13** is attached to the one end portions of the coaxial cables and is mechanically and electrically coupled to the mating connector device, FIG. **18**B is a cross-sectional view taken along line B-B of FIG. **18**A, FIG. **18**C is a cross-sectional view taken along line C-C of FIG. **18**A, and FIG. **18**D is a cross-sectional view taken along line D-D of FIG. **18**A.

[0041] FIGS. **19**A to **19**D are views for describing a method for assembling the coaxial connector device of FIG. **13**.

[0042] FIGS. **20**A and **20**B are views for describing the method for assembling the coaxial connector device of FIG. **13**, and are views showing steps following FIG. **19**D.

DESCRIPTION OF EMBODIMENTS

[0043] FIG. **1** is a perspective view showing a coaxial connector device **11** (electrical connector) according to the present embodiment. The coaxial connector device **11** shown in FIG. **1** is used by being attached to one end portions of two coaxial cables. The coaxial cables in which the coaxial connector device **11** is attached to the one end portions are shown as coaxial cables **12**A and **12**B in FIG. **4** to be described later. Each of the coaxial cables **12**A and **12**B includes a center conductor **13**; an inner insulator **14** that tightly surrounds the center conductor **13**; an outer conductor **15** that tightly surrounds the inner insulator **14**; and a skin insulator **16** that tightly surrounds the outer conductor **15**. At each of the one end portions of the coaxial cables **12**A and **12**B to which the coaxial connector device **11** is attached, the skin insulator **16** is partially cut away to expose the outer conductor **15**, and each of the outer conductor **15** and the inner insulator **14** is partially cut away to expose the center conductor **13**.

[0044] FIG. **2** is an exploded perspective view showing the coaxial connector device **11** according to the present embodiment. As shown in FIGS. **1** and **2**, the coaxial connector device **11** includes, as main components, two signal contact members **20**A and **20**B (a first signal contact member and a second signal contact member), a ground contact member **30**, and a conductive member **50** (refer to FIG. **2**). In addition, the coaxial connector device **11** may include an insulating housing member **40**.

[0045] The insulating housing member **40** is made of an insulating material such as a synthetic resin material. The insulating housing member **40** supports the signal contact members **20**A and **20**B and the ground contact member **30** in a mutually insulated state. The insulating housing member **40** includes a first supporting portion **41** related to supporting the signal contact member **20**A, a second supporting portion **42** related to supporting the signal contact member **20**B, and a base portion **43** provided between the first supporting portion **41** and the second supporting portion **42** (therebetween in a width direction of the coaxial connector device **11**).

[0046] The first supporting portion **41** includes a tubular portion **44** (refer to FIG. **2**), a center conductor supporting portion **45** (refer to FIG. **2**), and a flat plate portion **46**. The tubular portion **44** holds the signal contact member **20**A. The center conductor supporting portion **45** extends from the tubular portion **44**, and supports the center conductor **13** of the coaxial cable **12**A to be connected to the signal contact member **20**A. The flat plate portion **46** extends along an inner surface of a base plate **34***a* (to be described later) of a shell portion **34** of the ground contact member **30**. The flat plate portion **46** comes into contact with a first portion **20***x* (to be described later) of the signal contact member **20**A to press a first portion **20***x* and to bend the first portion **20***x* in a state where the shell portion **34** has assumed a bent position (state of FIGS. **5** and **6**). A configuration of the second supporting portion **42** is same as the configuration of the first supporting portion **41** (the signal contact member **20**A and the coaxial cable **12**A are replaced with the signal contact member **20**B and the coaxial cable **12**B, and the same in that the tubular portion **44**, the center conductor supporting portion **45**, and the flat plate portion **46** are provided).

[0047] The ground contact member **30** is made of an elastic conductive material. A ground potential is applied to the ground contact member **30**, and the ground contact member **30** is electrically connected to the outer conductors **15** of the coaxial cables **12**A and **12**B. The ground contact member **30** includes an annular fitting portion **31**; a first cable supporting portion **32** related to supporting the coaxial cable **12**A; a second cable supporting portion **33** related to supporting the coaxial cable **12**B; and the shell portion **34**.

[0048] The annular fitting portion **31** is a portion that partially surrounds a part of the insulating housing member **40** (specifically, a portion including the tubular portions **44** of the first supporting portion **41** and of the second supporting portion **42**), and that is fitted and connected to an annular fitting portion **102** of a ground contact portion **101** of a mating connector device **100** (refer to FIG. **7**). An opening portion OP**1** (refer to FIG. **6**) interposed between a pair of facing end portions **31***a* and **31***a* of the annular fitting portion **31** is formed in the annular fitting portion **31**.

[0049] The first cable supporting portion **32** includes an arm portion **35**, a placement portion **36**, and a wall portion **37**. The arm portion **35** extends from one end portion **31***a* of the pair of end portions **31***a* of the annular fitting portion **31** along the center conductor supporting portion **45** of the insulating housing member **40**. The placement portion **36** is continuous with a lower end of the arm portion **35**, and extends in a horizontal direction (specifically, the width direction of the coaxial connector device **11**) such that the center conductor supporting portion **45** and the coaxial cable **12**A are placed thereon. The wall portion **37** is provided to be continuous with the placement portion **36** and to face the arm portion **35**. A protrusion **36***a* is provided on a placement surface of the placement portion **36** for the coaxial cable **12**A. The protrusion **36***a* is a projection-shaped portion that comes into contact with the outer conductor **15** of the coaxial cable **12**A (refer to FIG. **10**). The coaxial cable **12**A is positioned by being fixed (supported) in a region defined by the arm portion **35**, the placement portion **36**, and the wall portion **37**. A configuration of the second cable supporting portion **33** is the same as the configuration of the first cable supporting portion **32** (the one end portion **31***a* and the coaxial cable **12**A are replaced with the other end portion **31***a* and the coaxial cable **12**B, and the same in that the arm portion **35**, the placement portion **36**, the protrusion **36***a*, and the wall portion **37** are provided).

[0050] The shell portion **34** is a portion that bendably extends from one end of the annular fitting portion **31**, and that is connected to the outer conductors **15** of the coaxial cables **12**A and **12**B when the shell portion **34** is folded. The shell portion **34** selectively assumes an upright position where the shell portion **34** is not folded with respect to the annular fitting portion **31** and the bent position where the shell portion **34** is folded with respect to the annular fitting portion **31**. The shell portion **34** assumes the upright position in FIGS. **1**, **2**, and **4**, and assumes the bent position in FIGS. **5** and **6** to be described later. For the description of the shell portion **34**, FIGS. **5** and **6** will also be referred to in addition to FIGS. **1** and **2**.

[0051] The shell portion **34** includes the base plate **34***a* covering an upper surface of the annular fitting portion **31**; a first fixing portion **38** related to fixing the coaxial cable **12**A; and a second fixing portion **39** related to fixing the coaxial cable **12**B. As shown in FIG. **5**, the base plate **34***a* extends from a location of connection to the annular fitting portion **31** to a region bordering outer surfaces of the coaxial cables **12**A and **12**B. The first fixing portion **38** includes clamping portions **34***b*, **34***c*, and **34***d* that are portions standing with respect to the base plate **34***a* in the bent position (refer to FIGS. **5** and **6**). The clamping portions **34***b*, **34***c*, and **34***d* are foldably configured, and are barrel portions that clamp and hold the coaxial cable **12**A.

[0052] The clamping portion **34***b* is folded to cover the placement portion **36**, thereby sandwiching the placement portion **36**, the conductive member **50**, the center conductor supporting portion **45**, and the coaxial cable **12**A between the clamping portion **34***b* and the base plate **34***a* to fix positions relative to each other. Specifically, the clamping portion **34***b* fixes the coaxial cable **12**A to the placement portion **36** such that the protrusion **36***a* of the placement portion **36** comes into contact with the outer conductor **15** of the coaxial cable **12**A. Accordingly, the ground contact member **30** and the outer conductor **15** can be reliably brought into contact with and electrically connected to each other. The clamping portion **34***c* is folded to cover the coaxial cable **12**A with the outer conductor **15** exposed, on a side closer to a rear end of the coaxial cable **12**A than the clamping portion **34***b*, thereby sandwiching the coaxial cable **12**A between the clamping portion **34***c* and the base plate **34***a* to fix the position of the coaxial cable **12**A. The clamping portion **34***d* is folded to cover the skin insulator **16**, on a side closer to the rear end of the coaxial cable **12**A than the clamping portion **34***c*, thereby sandwiching the coaxial cable **12**A between the clamping portion **34***d* and the base plate **34***a* to fix the position of the coaxial cable **12**A. The coaxial cable **12**A with the outer conductor **15** exposed is clamped by the clamping portion **34***c*, so that the ground contact member **30** and the outer conductor **15** can be reliably brought into contact with and electrically connected to each other. A configuration of the second fixing portion **39** is the same as the configuration of the first fixing portion **38**

(the coaxial cable **12**A is replaced with the coaxial cable **12**B, and the same in that the clamping portions **34***b*, **34***c*, and **34***d* are provided).

[0053] The signal contact members **20**A and **20**B are made of an elastic conductive material. The signal contact member **20**A is connected to the center conductor **13** of the coaxial cable **12**A, and is connected to a signal contact portion **104** (refer to FIG. **7**) of the mating connector device **100** (mating connector) mounted on a circuit board (not shown). The signal contact member **20**B is connected to the center conductor **13** of the coaxial cable **12**B, and is connected to a signal contact portion **105** (refer to FIG. **7**) of the mating connector device **100** (mating connector) mounted on the circuit board (not shown).

[0054] As shown in FIGS. **2** and **10** and the like, the signal contact member **20**A includes the first portion **20***x*, a second portion **20***y*, a third portion **20***z*, and locking portions **20***w*. The signal contact member **20**A is disposed in the tubular portion **44** (refer to FIG. **10**). The second portion **20***y* is a portion extending in an extending direction of the coaxial cable **12**A, and is a portion that comes into contact with the center conductor **13** of the coaxial cable **12**A through an upper surface of the second portion **20***y*. The first portion **20***x* is a portion that bendably extends from a tip (end portion on a side away from the coaxial cable **12**A) of the second portion **20***y*. In a state where the shell portion **34** assumes the bent position (state of FIG. **10**), the first portion **20***x* is pressed by the flat plate portion **46**, and is bent to sandwich the center conductor **13** between the first portion **20***x* and the second portion **20***y*. The third portion **20***z* is a portion formed in a U shape that extends downward in a tip direction of the second portion **20***y* so as to be folded from a rear end (end portion opposite a side that is continuous with the first portion **20***x*) of the second portion **20***y*, and that extends downward, is folded at a lower end, and extends upward. The third portion **20***z* is connected to the signal contact portion **104** of the mating connector device **100** at a portion formed in a U shape (refer to FIG. **10**). The locking portions **20***w* are portions that are fixed to the insulating housing member **40** by engaging locking portions (not shown) of the insulating housing member **40**. The locking portions **20***w* extend downward from both side surfaces of the second portion **20***y*. A configuration of the signal contact member **20**B is same as the configuration of the signal contact member **20**A (the coaxial cable **12**A is replaced with the coaxial cable **12**B, and the same in that the first portion **20***x*, the second portion **20***y*, the third portion **20***z*, and the locking portions **20***w* are provided).

[0055] FIG. **3** is a perspective view of the conductive member **50** of the coaxial connector device **11** according to the present embodiment when viewed from above. As shown in FIG. **3**, the conductive member **50** includes a first region **51**, a second region **52**, an intermediate region **53**, an isolation characteristic improving portion **54** (a second ground connection portion and a second connection portion), and connection portions **59**.

[0056] The first region **51** includes a rear ground connection portion **55** (first ground connection portion) and a front ground connection portion **56** (the second ground connection portion and a first connection portion) that is continuous with the rear ground connection portion **55**. The rear ground connection portion **55** is a flat plate-shaped portion that is connected to (comes into contact with) the ground contact member **30**, specifically, the placement portion **36** of the first cable supporting portion **32** (refer to FIG. **10**).

[0057] The front ground connection portion **56** is a flat plate-shaped portion that is connected to (comes into contact with) the annular fitting portion **102** of the ground contact portion **101** of the mating connector device **100** (refer to FIG. **10**). When viewed in a direction of fitting to the mating connector device **100**, as shown in FIGS. **6** and **10**, the front ground connection portion **56** overlaps at least a part of a signal line including the center conductor **13** of the coaxial cable **12**A and the signal contact member **20**A (specifically, a part of the center conductor **13**). When viewed in the direction of fitting described above, as shown in FIG. **6**, the front ground connection portion **56** overlaps the clamping portion **34***b* that is a barrel portion, in an arrangement direction of the signal contact members **20**A and **20**B (in the width direction of the coaxial connector device **11**). In addition, as shown in FIG. **6**, the front ground connection portion **56** is disposed in the opening portion OP**1** interposed between the end portions **31***a* and **31***a* of the annular fitting portion **31**.

[0058] A configuration of the second region **52** is the same as the configuration of the first region **51** (the coaxial cable **12**A is replaced with the coaxial cable **12**B, and the same in that the rear ground connection portion **55** and the front ground connection portion **56** are provided). Namely, the conductive member **50** includes a pair of the front ground connection portions **56** and **56** to be connected to the annular fitting portion **102** of the mating connector device **100**, one front ground connection portion **56** (the front ground connection portion **56** of the first region **51**) overlaps a part of the center conductor **13** of the coaxial cable **12**A when viewed in the direction of fitting, and the other front ground connection portion **56** (the front ground connection portion **56** of the second region **52**) overlaps a part of the center conductor **13** of the coaxial cable **12**B when viewed in the direction of fitting.

[0059] The intermediate region **53** is a portion that extends in the arrangement direction of the signal contact members **20**A and **20**B (in the width direction of the coaxial connector device **11**) to couple the first region **51** and the second region **52** to each other. The respective connection portions **59** are provided to be continuous with the first region **51** and the second region **52**, and are portions extending upward. The connection portions **59** are portions that are inserted into the insulating housing member **40**. The connection portions **59** are inserted into respective insertion openings **49** (refer to FIG. **11**A) of the insulating housing member **40**, so that the conductive member **50** is fixed to the insulating housing member **40**.

[0060] As shown in FIGS. **3** and **9**, the isolation characteristic improving portion **54** includes an extension portion **57** extending from the intermediate region **53** in the extending direction of the coaxial cables **12**A and **12**B (tip direction of the coaxial cables **12**A and **12**B), and a contact portion **58** formed in a U shape to extend downward from a tip of the extension portion **57**, to be folded back at a lower end, and to extend upward As shown in FIG. **9**, the contact portion **58** is connected to (comes into contact with) a contact portion **103** of the ground contact portion **101** disposed between the signal contact portions **104** and **105** of the mating connector device **100**. Incidentally, a part of the conductive member **50** is connected to the clamping portions **34***b* that are barrel portions to clamp and hold the coaxial cables **12**A and **12**B, respectively, through the placement portions **36**, but may be directly connected to the clamping portions **34***b* that are barrel portions.

[0061] FIG. **4** is a perspective view showing a process of attaching the coaxial connector device **11** according to the present embodiment to the one end portions of the coaxial cables **12**A and **12**B. When the coaxial connector device **11** is attached to the one end portions of the coaxial cables **12**A and **12**B, first, as shown in FIG. **2**, the center conductors **13** of the coaxial cables **12**A and **12**B are placed on the respective second portions **20***y* of the signal contact members **20**A and **20**B. Each of the placed coaxial cables **12**A and **12**B include the one end portion that is in a state where the skin insulator **16** is partially cut away to expose the outer conductor **15** and each of the outer conductor **15** and the inner insulator **14** is partially cut away to expose the center conductor **13**.

[0062] Thereafter, the shell portion **34** of the ground contact member **30** that has assumed the upright position is folded with respect to the annular fitting portion **31** of the ground contact member **30** to assume the bent position. Therefore, the shell portion **34** bends the first portions **20***x* through the flat plate portions **46** of the insulating housing member **40**, and sandwiches the center conductors **13** between the first portions **20***x* and the second portions **20***y*. Accordingly, the center conductors **13** of the coaxial cables **12**A and **12**B are connected to the respective signal contact members **20**A and **20**B. In addition, the shell portion **34** abuts the outer conductors **15** of the coaxial cables **12**A and **12**B.

[0063] Further, thereafter, a pair of the clamping portions **34***b* and **34***b* of the shell portion **34** that has assumed the bent position are folded to clamp the respective placement portions **36** so as to wrap the placement portions **36** from outside, and to clamp the respective outer conductors **15** of the coaxial cables **12**A and **12**B. In addition, a pair of the clamping portions **34***c* and **34***c* and a pair of the clamping portions **34***d* and **34***d* are folded to clamp the respective outer conductors **15** of and the respective skin insulators **16** of the coaxial cables **12**A and **12**B. Accordingly, the shell portion **34** of the ground contact member **30** is connected and fixed to the outer conductors **15** of the coaxial cables **12**A and **12**B.

[0064] As a result, as shown in FIGS. **5** and **6**, a state where the coaxial connector device **11** is attached to the one end portions of the coaxial cables **12**A and **12**B is firmly maintained. FIG. **5** is a perspective view from above showing a state where the coaxial connector device **11** according to the present embodiment is attached to the one end portions of the coaxial cables **12**A and **12**B. FIG. **6** is a bottom view showing a state where the coaxial connector device **11** according to the present embodiment is attached to the one end portions of the coaxial cables **12**A and **12**B.

[0065] FIG. **7** is a perspective view showing the mating connector device **100** to which the coaxial connector device **11** according to the present embodiment is coupled. The mating connector device **100** is fixed onto a mounting surface (not shown) of the circuit board on which components and the like are mounted.

[0066] The mating connector device **100** includes the signal contact portions **104** and **105** (a first signal contact portion and a second signal contact portion) and the ground contact portion **101**. The signal contact portions **104** and **105** are made of a conductive material, and are electrically connected to respective signal terminals (not shown) provided on the mounting surface of the circuit board. The ground contact portion **101** is electrically connected to a ground potential portion (not shown) provided on the mounting surface of the circuit board. The ground contact portion **101** is formed as an annular body to surround the signal contact portions **104** and **105**. The ground contact portion **101** includes the annular fitting portion **102** (first portion) and the contact portion **103** (second portion). The annular fitting portion **102** is fitted and connected to the annular fitting portion **31** of the ground contact member **30**. The contact portion **103** is disposed between the signal contact portions **104** and **105**, and comes into contact with (is connected to) the contact portion **58** of the isolation characteristic improving portion **54**.

[0067] FIG. **8** is a plan view showing a state where the coaxial connector device **11** according to the present embodiment is attached to the one end portions of the coaxial cables **12**A and **12**B and is mechanically and electrically coupled to the mating connector device **100**. FIG. **9** is a cross-sectional view showing a cross section taken along line IX-IX in FIG. **8**. FIG. **10** is a cross-sectional view showing a cross section taken along line X-X in FIG. **8**. As shown in FIGS. **9** and **10**, in a state where the coaxial connector device **11** is coupled to the mating connector device **100**, the annular fitting portion **31** of the ground contact member **30** in the coaxial connector device **11** is fitted and connected to the annular fitting portion **102** of the ground contact portion **101** of the mating connector device **100**.

[0068] In a fitted and connected state, as shown in FIG. **10**, the third portion **20***z* of the signal contact member **20**A connected to the center conductor **13** of the coaxial cable **12**A is brought into contact with (connected to) the signal contact portion **104** of the mating connector device **100**. Similarly, the third portion **20***z* of the signal contact member **20**B connected to the center conductor **13** of the coaxial cable **12**B is brought into contact with (connected to) the signal contact portion **105** of the mating connector device **100**. Accordingly, the center conductors **13** of the coaxial cables **12**A and **12**B are electrically connected to the respective signal terminals (not shown) provided on the mounting surface of the circuit board, through the signal contact members **20**A and **20**B of the coaxial connector device **11** and through the signal contact portions **104** and **105** of the mating connector device **100**.

[0069] In addition, in a fitted and connected state, as shown in FIG. **10**, the conductive member **50** is in contact with the placement portion **36** of which the protrusion **36***a* is in contact with the outer conductor **15** of the coaxial cable **12**A, at the rear ground connection portion **55** of the first region **51**, and is in contact with the annular fitting portion **102** of the ground contact portion **101** of the mating connector device **100** at the front ground connection portion **56** of the first region **51**. Similarly, the conductive member **50** is in contact with the placement portion **36** of which the protrusion **36***a* is in contact with the outer conductor **15** of the coaxial cable **12**B, at the rear ground connection portion **55** of the second region **52**, and is in contact with the annular fitting portion **102** of the ground contact portion **101** of the mating connector device **100** at the front ground connection portion **56** of the second region **52**. Accordingly, the outer conductors **15** of the coaxial cables **12**A and **12**B are electrically connected to the ground potential portion (not shown) provided on the circuit board, through the ground contact member **30** of and through the conductive member **50** of the coaxial connector device **11** and through the ground contact portion **101** of the mating connector device **100**.

[0070] Further, in a fitted and connected state, as shown in FIG. **9**, the contact portion **58** of the isolation characteristic improving portion **54** of the conductive member **50** is in contact with the contact portion **103** of the ground contact portion **101** disposed between the signal contact portions **104** and **105** of the mating connector device **100**. Accordingly, the isolation characteristic improving portion **54** electrically connected to the ground potential portion (not shown) provided on the circuit board is provided between signal lines related to the coaxial cables **12**A and **12**B.

[0071] Next, a method for assembling the coaxial connector device **11** according to the present embodiment will be described with reference to FIGS. **11** and **12**. FIGS. **11** and **12** are views for describing the method for assembling the coaxial connector device according to the present embodiment, and shows assembly steps in order of FIGS. **11**A, **11**B, **11**C, **11**D, **12**A, **12**B, and **12**C.

[0072] In an initial step, as shown in FIG. **11**A, the conductive member **50** and the insulating housing member **40** are prepared, and as shown in FIG. **11**B, the connection portions **59** of the conductive member **50** are inserted into the respective insertion openings **49** of the insulating housing member **40**, so that the conductive member **50** and the insulating housing member **40** are integrated.

[0073] Subsequently, as shown in FIG. **11**C, the insulating housing member **40** with which the conductive member **50** is integrated is assembled to the ground contact member **30**. Specifically, the insulating housing member **40** is assembled to the ground contact member **30** such that the tubular portions **44** of the insulating housing member **40** are surrounded by the annular fitting portion **31** of the ground contact member **30** and the center conductor supporting portions **45** of the insulating housing member **40** are placed on the respective placement portions **36** of the ground contact member **30**.

[0074] Subsequently, as shown in FIG. **11**D, the signal contact members **20**A and **20**B are placed in the respective tubular portions **44** of the insulating housing member **40** assembled to the ground contact member **30**. The coaxial connector device **11** shown in FIG. **12**A in which the ground contact member **30** assumes the upright position is prepared by the assembly steps up to this point.

[0075] Subsequently, as shown in FIG. **12**B, the coaxial connector device **11** is attached to the one end portions of the coaxial cables **12**A and **12**B. Specifically, the center conductors **13** of the coaxial cables **12**A and **12**B are placed on the respective second portions **20***y* (refer to FIG. **4**) of the signal contact members **20**A and **20**B.

[0076] Therefore, the shell portion **34** of the ground contact member **30** that has assumed the upright position comes to assume the bent position, and the clamping portions **34***b*, **34***c*, and **34***d* are folded, so that the center conductors **13** of the coaxial cables **12**A and **12**B are connected to the respective signal contact members **20**A and **20**B. In addition, the shell portion **34** of the ground contact member **30** is connected and fixed to the outer conductors **15** of the coaxial cables **12**A and **12**B. As a result, as shown in FIG. **12**C, a state where the coaxial connector device **11** is attached to the one end portions of the coaxial cables **12**A and **12**B is firmly maintained.

[0077] Next, actions and effects of the coaxial connector device **11** according to the present embodiment will be described.

[0078] A coaxial connector device **11** according to the present embodiment includes a signal contact member **20**A that is connected to a center conductor **13** of a coaxial cable **12**A, and that is connected to a signal contact portion **104** of a mating connector device **100** mounted on a circuit board; a signal contact member **20**B that is connected to the center conductor **13** of a coaxial cable **12**B, and that is connected to a signal contact portion **105** of the mating connector device **100**; a ground contact member **30** which includes an annular fitting portion **31** that is fitted and connected to an annular fitting portion **102** of a ground contact portion **101** of the mating connector device, and a shell portion **34** that bendably extends from the annular fitting portion **31** to be connected to outer conductors **15** of the coaxial cable **12**A and of the coaxial cable **12**B, and to which a ground potential is applied; and a conductive member **50** including a rear ground connection portion **55** that is connected to the ground contact member **30**, and a front ground connection portion **56** that is connected to the ground contact portion **101** of the mating connector device **100**. When viewed in a direction of fitting to the mating connector device **100**, the front ground connection portion **56** of the conductive member **50** overlaps at least a part of a signal line including the center conductor **13** of the coaxial cable **12**A and the signal contact member **20**A (specifically, a part of the center conductor **13**), and at least a part of a signal line including the center conductor **13** of the coaxial cable **12**B and the signal contact member **20**B (specifically, a part of the center conductor **13**).

[0079] The coaxial connector device **11** according to the present embodiment includes the signal contact member **20**A connected to the center conductor **13** of the coaxial cable **12**A; the signal contact member **20**B connected to the center conductor **13** of the coaxial cable **12**B; the ground contact member **30** to which a ground potential is applied; and the conductive member **50**. The rear ground connection portion **55** of the conductive member **50** is connected to the ground contact member **30**, and the front ground connection portion **56** is connected to the ground contact portion **101** of the mating connector device **100**. Furthermore, in the coaxial connector device **11**, when viewed in the direction of fitting, the front ground connection portion **56** overlaps at least a part of the signal line including the center conductor **13** of the coaxial cable **12**A and the signal contact member **20**A, and at least a part of the signal line including the center conductor **13** of the coaxial cable **12**B and the signal contact member **20**B. Since the front ground connection portion **56** overlaps at least a part of these signal lines, signals of these signal lines are prevented from propagating to the outside (surroundings of the coaxial connector device **11**). As a result, signals of the signal lines can be prevented from becoming noise and affecting surrounding external products, EMI characteristics of the coaxial connector device **11** can be improved, and noise resistance can be improved.

[0080] The shell portion **34** of the ground contact member **30** may include clamping portions **34***b* (barrel portions) that clamp and hold the coaxial cable **12**A and the coaxial cable **12**B, respectively, and the conductive member **50** may be connected to the clamping portions **34***b*. In such a manner, since the clamping portions **34***b* that hold the coaxial cables **12**A and **12**B forming the signal lines and the conductive member **50** are connected to (in contact with) each other, contact locations can be reliably set to a ground potential, and portions at which the signal lines and the conductive member **50** (specifically, the front ground connection portion **56**) overlap each other can be suitably formed.

[0081] The ground contact portion **101** of the mating connector device **100** may include the annular fitting portion **102** that is fitted and connected to the annular fitting portion **31** of the ground contact member **30**. The conductive member **50** may include a pair of the front ground connection portions **56** that are connected to the annular fitting portion **102**. One of the pair of front ground connection portions **56** may overlap at least a part of one signal line when viewed in the direction of fitting, and the other of the pair of front ground connection portions **56** may overlap at least a part of the other signal line when viewed in the direction of fitting. In such a manner, since the pair of front ground connection portions **56** are connected to a portion related to fitting (annular fitting portion **102**) in the ground contact portion **101** of the mating connector device **100**, in a state where the pair of front ground connection portions **56** are fitted to the mating connector device **100**, a contact location can be reliably set to a ground potential. Furthermore, since one of the pair of front ground connection portions **56** overlaps at least a part of the one signal line, and the other of the pair of front ground connection portions **56** overlaps at least a part of the other signal line, signals of both the signal lines can be reliably prevented from propagating to the outside (surroundings of the coaxial connector device **11**).

[0082] The shell portion **34** of the ground contact member **30** may include clamping portions **34***b* that clamp and hold the coaxial cable **12**A and the coaxial cable **12**B, respectively. When viewed in the direction of fitting, the pair of front ground connection portions **56** may overlap the clamping portions **34***b* in an arrangement direction of the signal contact members **20**A and **20**B. In such a manner, since the pair of front ground connection portions **56** and the clamping portions **34***b* that hold the coaxial cables **12**A and **12**B overlap each other, portions at which the pair of front ground connection portions **56** and the signal lines (specifically, the center conductors **13** of the coaxial cables **12**A and **12**B) overlap each other can be suitably formed. As a result, signals of both the signal lines can be appropriately prevented from propagating to the outside (surroundings of the coaxial connector device **11**).

[0083] An opening portion OP**1** may be formed in the annular fitting portion **31** of the ground contact member **30**, and the pair of front ground connection portions **56** may be disposed in the opening portion OP**1**. In such a manner, for example, since the pair of front ground connection portions **56** forming the conductive member **50** are provided in the opening portion OP**1** that is formed in a process of manufacturing, the opening portion OP**1** is partially closed by the pair of front ground connection portions **56**, so that signals (high-frequency signals) of the signal lines can be effectively prevented from leaking from the opening portion OP**1** to the outside.

[0084] The ground contact portion **101** of the mating connector device **100** may include a contact portion **103** disposed between the signal contact portion **104** and the signal contact portion **105**, and the conductive member **50** may include an isolation characteristic improving portion **54** that is connected to the contact portion **103**. Since the isolation characteristic improving portion **54** is connected to the contact portion **103** of the ground contact portion **101** of the mating connector device **100** disposed between the signal contact portion **104** and the signal contact portion **105**, signals are prevented from propagating between the signal contact members **20**A and **20**B that are different from each other. As a result, an isolation characteristic can be improved, and noise resistance can be improved.

[0085] An aspect according to the present invention is not limited to the above embodiment. FIGS. **13** and **14** are exploded perspective views showing a coaxial connector device **511** according to another aspect of the present embodiment. FIG. **15** is a perspective view from below showing a state where the coaxial connector device **511** is attached to one end portions of coaxial cables. Hereinafter, among configurations of the coaxial connector device **511**, configuration different from those of the coaxial connector device **11** according to the above embodiment will be mainly described, and a description of configurations common to (corresponding to) the coaxial connector device **11** may not be repeated.

[0086] As shown in FIGS. **13** to **15**, the coaxial connector device **511** includes, as main components, two signal contact members **520**A and **520**B (a first signal contact member and a second signal contact member), a ground contact member **530**, and a conductive member **550**. In addition, the coaxial connector device **511** may include an insulating housing **540**.

[0087] The insulating housing **540** supports the signal contact members **520**A and **520**B and the ground contact member **530** in a mutually insulated state. The insulating housing **540** includes a first supporting portion **541** related to supporting the signal contact member **520**A (refer to FIG. **19**C); a second supporting portion **542** related to supporting the signal contact member **520**B (refer to FIG. **19**C); and a base portion **543**. The first supporting portion **541** and the second supporting portion **542** are disposed adjacent to each other in a width direction of the coaxial connector device **511**.

[0088] Protrusion portions **591** protruding outward in the width direction are formed on respective outer surfaces of the first supporting portion **541** and of the second supporting portion **542** in the width direction. The protrusion portions **591** are portions that are fitted to respective recessed portions **535***x* (details will be described later) of arm portions **535** of the ground contact member **530** to fix a position of the insulating housing **540** with respect to the ground contact member **530** when the coaxial connector device **511** is assembled (refer to FIGS. **19**C and **19**D).

[0089] As shown in FIG. **14**, insertion openings **549** that are recessed portions are formed in both respective end portions of lower surfaces of the first supporting portion **541** and of the second supporting portion **542** in the width direction. The insertion openings **549** are portions into which respective connection portions **559** (details will be described later) of the conductive member **550** are inserted. The connection portions **559** are inserted into the respective insertion openings **549** to fix a position of the conductive member **550** with respect to the insulating housing **540**.

[0090] As shown in FIG. **13**, the base portion **543** is a portion that is continuous with the first supporting portion

541 and with the second supporting portion 542 in the extending direction of the coaxial cables 12A and 12B, and is a portion located closer to the tip direction (front direction) of the coaxial cables 12A and 12B than the first supporting portion 541 and the second supporting portion 542. A recessed portion 546 for fixing an isolation characteristic improving portion 570 to be described later is formed in a central portion of the base portion 543 in the width direction of the coaxial connector device 511. The recessed portion 546 is formed along the extending direction of the coaxial cables 12A and 12B from a portion interposed between the first supporting portion 541 and the second supporting portion 542 to a tip portion of the base portion 543.

[0091] As shown in FIG. 14, contact fixing portions 547 and 548 extending downward are formed at respective tip portions of a lower surface of the base portion 543. The contact fixing portion 547 is a portion around which a third portion 520*z* (details will be described later) of the signal contact member 520A extending to penetrate through an inside of the base portion 543 is wound, and to which the third portion 520*z* is fixed. The contact fixing portion 548 is a portion around which the third portion 520*z* (details will be described later) of the signal contact member 520B extending to penetrate through the inside of the base portion 543 is wound, and to which the third portion 520*z* is fixed (refer to FIG. 18C).

[0092] A ground potential is applied to the ground contact member 530, and the ground contact member 530 is electrically connected to the outer conductors 15 of the coaxial cables 12A and 12B through a ground bar 581 (details will be described later) (refer to FIG. 18C). The ground contact member 530 includes an annular fitting portion 531, a pair of the arm portions 535 and 535, and a shell portion 534.

[0093] The annular fitting portion 531 is a portion that partially surrounds a part of the insulating housing 540 (specifically, the base portion 543), and that is fitted and connected to an annular fitting portion 630 forming a ground contact portion of a mating connector device 600 (refer to FIG. 16). The annular fitting portion 531 is formed in a substantially cylindrical shape, and a rear end portion thereof is not continuous, and an opening portion 595 is formed in the annular fitting portion 531 (refer to FIG. 15).

[0094] The pair of arm portions 535 and 535 are portions that are continuous with the annular fitting portion 531, and that extend along the extending direction of the coaxial cables 12A and 12B while facing each other in the width direction of the coaxial connector device 511. One arm portion 535 extends along an outer surface of the first supporting portion 541 in the extending direction. The other arm portion 535 extends along an outer surface of the second supporting portion 542 in the extending direction. The recessed portion 535*x* that is fitted to the protrusion portion 591 of the insulating housing 540 is formed in each of the pair of arm portions 535 and 535.

[0095] The shell portion 534 includes a base plate 534*a* covering an upper surface of the annular fitting portion 531; a first fixing portion 538 (barrel portion) related to fixing the coaxial cable 12A; and a second fixing portion 539 (barrel portion) related to fixing the coaxial cable 12B. The base plate 534*a* is continuous with a tip of the annular fitting portion 531, and extends rearward to a position where the base plate 534*a* covers the upper ground bar 581 (details will be described later) of a cable fixing portion 580 (details will be described later), at the bent position.

[0096] The first fixing portion 538 holds the coaxial cable 12A. The first fixing portion 538 is a portion that extends continuously from an outer edge portion of a rear end portion (portion behind a portion of the base plate 534*a* covering the upper surface of the annular fitting portion 531) of the base plate 534*a* in the width direction. The first fixing portion 538 is foldably configured, and includes a clamping portion 534*b* covering the arm portion 535 from outside in the width direction, and a clamping portion 534*c* covering the cable fixing portion 580 (lower ground bar 581 to be described in detail later) from below, at the bent position. The clamping portion 534*b* is folded at a location where the clamping portion 534*b* is continuous with the base plate 534*a*, to cover the arm portion 535. The clamping portion 534*c* is folded at a location where the clamping portion 534*c* is continuous with the clamping portion 534*b*, to cover the lower ground bar 581. In such a manner, the first fixing portion 538 covers and holds the cable fixing portion 580 instead of directly clamping the coaxial cable 12A, to hold the coaxial cable 12A fixed by the cable fixing portion 580. A configuration of the second fixing portion 539 is the same as the configuration of the first fixing portion 538 (the coaxial cable 12A is replaced with the coaxial cable 12, and the same in that the clamping portions 534*b* and 534*c* are provided). Namely, the second fixing portion 539 covers and holds the cable fixing portion 580 instead of directly clamping the coaxial cable 12B, to hold the coaxial cable 12B fixed by the cable fixing portion 580.

[0097] The cable fixing portion 580 is configured to fix the coaxial cables 12A and 12B. The cable fixing portion 580 includes the ground bars 581 and 581 each having a plate shape and facing each other in an up-down direction, and a solder portion 582 that fixes the coaxial cables 12A and 12B by filling a gap between the ground bars 581 and 581 and the coaxial cables 12A and 12B therewith. The filling of the solder portion 582 is performed to cover the outer conductors 15, and the ground bars 581 and 581 cover an upper surface of and a lower surface of the solder portion 582, so that the ground bars 581 and 581 and the outer conductors 15 are electrically connected to each other.

[0098] A tip portion of a central portion of the upper ground bar 581 in the width direction is folded in a down direction, passes through an inside of the solder portion 582, and is connected to the isolation characteristic improving portion 570 (details will be described later) (refer to FIG. 18B). In such a manner, one ground bar 581 and the isolation characteristic improving portion 570 are in contact with each other, so that the isolation characteristic improving portion 570 is electrically connected to the outer conductors 15 of the coaxial cables 12A and 12B through the ground bar 581.

[0099] The signal contact member 520A is a conductive member that is connected to the center conductor 13 of the coaxial cable 12A, and that is connected to a signal contact portion 604 (refer to FIG. 16) of the mating connector device 600. The signal contact member 520B is connected to the center conductor 13 of the coaxial cable 12B, and is connected to a signal contact portion 605 (refer to FIG. 16) of the mating connector device 600.

[0100] The signal contact member 520A includes a first portion 520*x*, a second portion 520*y*, and the third portion 520*z*. The signal contact member 520A is supported by the first supporting portion 541. The second portion 520*y* is a portion extending in the extending direction of the coaxial cable **12**A, and is a portion that comes into contact with the center conductor **13** of the coaxial cable **12**A through an upper surface of the second portion **520***y*. The first portion **520***x* is a portion that is continuous with a tip of the second portion **520***y* and that extends forward. The first portion **520***x* includes a portion that is continuous with the tip of the second portion **520***y* and that extends to be inclined in an up direction, and a portion that is continuous with the portion extending to be inclined and that extends horizontally. The third portion **520***z* is a portion formed in a U shape that is continuous with a tip of the first portion **520***x*, and that extends downward, is folded at a lower end of a portion extending downward, and extends upward. The third portion **520***z* is connected to the signal contact portion **604** (refer to FIG. **16**) of the mating connector device **600** at a portion formed in a U shape. A configuration of the signal contact member **520**B is same as the configuration of the signal contact member **520**A (the coaxial cable **12**A is replaced with the coaxial cable **12**B, and the same in that the first portion **520***x*, the second portion **520***y*, and the third portion **520***z* are provided).

[0101] The conductive member **550** includes a rear ground connection portion **555** (first ground connection portion); an inclined portion **556** that is continuous with the rear ground connection portion **555**; and a front ground connection portion **557** (a second ground connection portion and a first connection portion) that is continuous with the inclined portion **556**.

[0102] The rear ground connection portion **555** is a flat plate-shaped portion that is connected to (comes into contact with) the ground contact member **530**, specifically, the clamping portions **534***c* of the first fixing portion **538** and of the second fixing portion **539** (refer to FIG. **18**C). A cutout **558** is formed in a central portion of a rear end of the rear ground connection portion **555** in the width direction. The cutout **558** is filled with solder, so that the lower ground bar **581** and the conductive member **550** are electrically connected to each other. Accordingly, an improvement in shieldability caused by reliable contact and mechanical connection caused by strong connection are ensured. Incidentally, as shown in FIG. **18**B, a pass-through portion **536** of the base plate **534***a* is filled with solder, so that the upper ground bar **581** and the ground contact member **530** are electrically connected to each other. Such soldering is performed after the caulking of the first fixing portion **538** (barrel portion) and the second fixing portion **539** that are barrel portions. The connection portions **559** protruding upward are provided at both respective end portions of a front end portion of the rear ground connection portion **555** in the width direction. The connection portions **559** are inserted into the respective insertion openings **549** of the insulating housing **540** to fix a position of the rear ground connection portion **555** with respect to the insulating housing **540**. The inclined portion **556** is a portion that is continuous with a front end of the rear ground connection portion **555** and that extends forward while being inclined, and has a shape inclined in an upward diagonal direction according to an inclined shape of a lower surface of the insulating housing **540**.

[0103] The front ground connection portion **557** is a flat plate-shaped portion that is connected to (comes into contact with) the annular fitting portion **630** (first portion) forming the ground contact portion of the mating connector device **100** (refer to FIG. **18**C). The front ground connection portion **557** is a portion that is continuous with a front end of the inclined portion **556**, and that extends forward while being inclined downward. A tip portion of the front ground connection portion **557** is disposed in the opening portion **595** (refer to FIG. **15**) of the annular fitting portion **531**, and the front ground connection portion **557** is connected to the annular fitting portion **630** at the tip portion thereof (refer to FIG. **18**C). The front ground connection portion **557** is continuously formed in the width direction that is an arrangement direction of the signal contact member **520**A and the signal contact member **520**B. Being continuously formed mentioned here refers to being formed without a cutout, a gap, or the like formed. Furthermore, when viewed in the direction of fitting, the front ground connection portion **557** overlaps at least a part (both sides in the present embodiment) of two signal lines.

[0104] Further, the conductive member **550** includes the isolation characteristic improving portion **570** (the second ground connection portion and a second connection portion) formed of a member separate from the plate-shaped member including the rear ground connection portion **555**, the inclined portion **556**, and the front ground connection portion **557** described above. In such a manner, the isolation characteristic improving portion **570** is formed of a member separate from other regions (front ground connection portion **557**) and the like in the second ground connection portion.

[0105] The isolation characteristic improving portion **570** is configured to extend in the extending direction of the signal contact member **520**A and of the signal contact member **520**B between the signal contact member **520**A and the signal contact member **520**B, and to prevent signals from propagating between the signal contact member **520**A and the signal contact member **520**B. The isolation characteristic improving portion **570** is a wall-shaped member having surfaces facing the respective signal contact members **520**A and **520**B. The isolation characteristic improving portion **570** is disposed between the signal contact member **520**A and the signal contact member **520**B to block at least a part of a region between the signal contact member **520**A and the signal contact member **520**B.

[0106] As shown in FIG. **18**B, the isolation characteristic improving portion **570** may be disposed between the signal contact member **520**A and the signal contact member **520**B to block an entirety of the region between the signal contact member **520**A and the signal contact member **520**B. As shown in FIGS. **18**B and **18**C, an area of the isolation characteristic improving portion **570** when viewed in the width direction is larger than an area of each of the signal contact member **520**A and the signal contact member **520**B when viewed in the width direction.

[0107] As shown in FIG. **18**B, the isolation characteristic improving portion **570** includes a first portion **570***x*, a second portion **570***y*, a third portion **570***z*, and a fourth portion **570***v*. The second portion **570***y* is a portion extending along the second portions **520***y* of the signal contact members **520**A and **520**B, and is connected to the ground bar **581** electrically connected to the outer conductors **15** of the coaxial cables **12**A and **12**B, through an upper surface of the second portion **570***y*. The first portion **570***x* is a portion extending along the first portions **520***x* of the signal contact members **520**A and **520**B. At least a part of an upper end portion of the first portion **570***x* is in contact with the base plate **534***a* of the shell portion **534** (refer to FIG. **18**B). The third portion **570***z* is a portion extending downward along the third portions **520**z of the signal contact members **520**A and **520**B. The third portion **570**z is connected to (comes into contact with) a contact portion **603** of the ground contact portion of the mating connector device **600**. The fourth portion **570**v is a portion that is continuous with the first portion **570**x and that extends downward, and is a portion that is press-fitted into the insulating housing **540**.

[0108] FIG. **16** is a perspective view showing the mating connector device **600** to which the coaxial connector device **511** is coupled. FIG. **17** is an exploded perspective view from above showing the mating connector device **600** of FIG. **16**.

[0109] As shown in FIGS. **16** and **17**, the mating connector device **600** includes the signal contact portions **604** and **605** (a first signal contact portion and a second signal contact portion), the annular fitting portion **630** and the contact portion **603** that form the ground contact portion, and an insulating housing **640**.

[0110] The insulating housing **640** supports the signal contact portions **604** and **605** and the annular fitting portion **630** and the contact portion **603** that form the ground contact portion, in a mutually insulated state. The signal contact portion **604** includes a pair of contact portions **604**a and **604**b. The contact portions **604**a and **604**b come into contact with (are connected to) the third portions **520**z of the signal contact member **520**A so as to sandwich the third portion **520**z therebetween. The signal contact portion **605** includes a pair of contact portions **605**a and **605**b. The contact portions **605**a and **605**b come into contact with (are connected to) the third portions **520**z of the signal contact member **520**B so as to sandwich the third portion **520**z therebetween.

[0111] The contact portion **603** includes a pair of contact portions **603**a and **603**b. The contact portions **603**a and **603**b come into contact with (are connected to) the third portion **570**z of the isolation characteristic improving portion **570** so as to sandwich the third portion **570**z therebetween. The annular fitting portion **630** is fitted and connected to the annular fitting portion **531** of the ground contact member **530**.

[0112] FIG. **18**A is a plan view showing a state where the coaxial connector device **511** is attached to the one end portions of the coaxial cables **12**A and **12**B and is mechanically and electrically coupled to the mating connector device **600**. FIG. **18**B is a cross-sectional view taken along line B-B of FIG. **18**A. FIG. **18**C is a cross-sectional view taken along line C-C of FIG. **18**A. FIG. **18**D is a cross-sectional view taken along line D-D of FIG. **18**A.

[0113] As shown in FIGS. **18**B and **18**C, in a state where the coaxial connector device **511** is coupled to the mating connector device **600**, the annular fitting portion **531** of the ground contact member **530** in the coaxial connector device **511** is fitted and connected to the annular fitting portion **630** of the ground contact portion of the mating connector device **600**.

[0114] In a fitted and connected state, as shown in FIG. **18**C, the third portion **520**z of the signal contact member **520**B to which the second portion **520**y and the center conductor **13** of the coaxial cable **12**B are connected comes into contact with the pair of contact portions **604**a and **604**b of the signal contact portion **604** of the mating connector device **600** (as for the signal contact member **520**A, comes into contact with the pair of contact portions **605**a and **605**b of the signal contact portion **605**). Accordingly, the center conductors **13** of the coaxial cables **12**A and **12**B are electrically connected to signal terminals (not shown) provided on a mounting surface of a circuit board through the signal contact members **520**A and **520**B of the coaxial connector device **511** and through the signal contact portions **604** and **605** of the mating connector device **600**.

[0115] In addition, in a fitted and connected state, as shown in FIG. **18**C, the rear ground connection portion **555** of the conductive member **550** is in contact with the clamping portions **534**c of the ground contact member **530** and with the ground bar **581** of the cable fixing portion **580**, and the front ground connection portion **557** is in contact with the annular fitting portion **630** forming the ground contact portion of the mating connector device **600**. The ground bar **581** is electrically connected to the outer conductors **15** of the coaxial cables **12**A and **12**B. Accordingly, the outer conductors **15** of the coaxial cables **12**A and **12**B are electrically connected to a ground potential portion (not shown) provided on the circuit board, through the ground contact member **530** of and through the conductive member **550** of the coaxial connector device **511** and through the annular fitting portion **630** of the mating connector device **600**.

[0116] Further, in a fitted and connected state, as shown in FIG. **18**B, the

[0117] isolation characteristic improving portion **570** connected to the ground bar **581** extends in the extending direction of the signal contact member **520**A and of the signal contact member **520**B between the signal contact member **520**A and the signal contact member **520**B, and is disposed between the signal contact member **520**A and the signal contact member **520**B. Furthermore, the third portion **570**z of the isolation characteristic improving portion **570** is in contact with the pair of contact portions **603**a and **603**b of the contact portion **603** of the ground contact portion of the mating connector device **600**. Accordingly, the isolation characteristic improving portion **570** is provided between the signal lines to prevent signals from propagating between the signal lines. Incidentally, the isolation characteristic improving portion **570** is electrically connected to the ground potential portion (not shown) provided on the circuit board.

[0118] Next, a method for assembling the coaxial connector device **511** will be described with reference to FIGS. **19** and **20**. FIGS. **19**A to **19**D are views for describing the method for assembling the coaxial connector device **511**. FIGS. **20**A and **20**B are views for describing the method for assembling the coaxial connector device **511**, and are views showing steps following FIG. **19**D.

[0119] In an initial step, as shown in FIG. **19**A, the insulating housing **540** with which the conductive member **550** and the signal contact members **520**A and **520**B are integrally molded, and the isolation characteristic improving portion **570** are prepared, and the connection portions **559** of the conductive member **550** are inserted into the insertion openings **549** of the insulating housing **540**, and the isolation characteristic improving portion **570** is fixed to the recessed portion **546** (refer to FIG. **13**) of the insulating housing **540**, so that as shown in FIG. **19**B, the conductive member **550** and the like are integrated with the insulating housing **540**.

[0120] Subsequently, as shown in FIG. **19**C, the insulating housing **540** with which the conductive member **550** and the like are integrated is assembled to the ground contact member **530**. Specifically, the insulating housing **540** is assembled to the ground contact member **530** such that the protrusion portions **591** of the insulating housing **540** are fitted to the recessed portions **535***x* of the pair of arm portions **535** and **535** (refer to FIG. **19**D).

[0121] Subsequently, as shown in FIG. **20**A, the coaxial connector device **511** is attached to the one end portions of the coaxial cables **12**A and **12**B. During attachment, the signal contact member **520**A and the center conductor **13** of the coaxial cable **12**A are connected to each other, and the signal contact member **520**B and the center conductor **13** of the coaxial cable **12**B are connected to each other. In addition, during attachment, the upper ground bar **581** and the isolation characteristic improving portion **570** are connected to each other. Thereafter, the shell portion **534** of the ground contact member **530** that has assumed the upright position comes to assume the bent position, and the clamping portion **534***b* and **534***c* are folded, so that as shown in FIG. **20**B, a state where the coaxial connector device **511** is attached to the one end portions of the coaxial cables **12**A and **12**B is firmly maintained. After the clamping portion **534***b* and **534***c* are folded, the cutout **558**, the pass-through portion **536**, and the like are filled with solder.

[0122] Next, actions and effects of the coaxial connector device **511** will be described.

[0123] In a coaxial connector device **511**, a shell portion **534** of a ground contact member **530** may include a first fixing portion **538** and a second fixing portion **539** as barrel portions that hold a coaxial cable **12**A and a coaxial cable **12**B, respectively, and a conductive member **550** may be connected to the first fixing portion **538** and to the second fixing portion **539**. As described above, even when the first fixing portion **538** and the second fixing portion **539** cover and hold the cable fixing portion **580** instead of directly clamping the coaxial cable **12**A and the coaxial cable **12**B, portions at which signal lines and the conductive member **550** overlap each other can be suitably formed.

[0124] A front ground connection portion **557** of the conductive member **550** may be connected to an annular fitting portion **630**, continuously formed in a width direction that is an arrangement direction of a signal contact member **520**A and a signal contact member **520**B, and overlap at least a part of two signal lines when viewed in a direction of fitting. In such a manner, since the front ground connection portion **557** is connected to a portion related to fitting (annular fitting portion **630**) in the ground contact portion of the mating connector device **600**, in a state where the front ground connection portion **557** is fitted to the mating connector device **600**, a contact location can be reliably set to a ground potential. Furthermore, since the front ground connection portion **557** that is continuously formed without a cutout, a gap, or the like formed overlaps at least a part of the two signal lines, signals of the two signal lines can be more appropriately prevented from propagating to the outside.

[0125] An opening portion **595** may be formed in the annular fitting portion **630**, and the front ground connection portion **557** may be formed in the opening portion **595**. In such a manner, for example, since the front ground connection portion **557** that is conductive member is provided in the opening portion **595** that is formed in a process of manufacturing, the opening portion **595** is effectively closed by the front ground connection portion **557** in which a cutout, a gap, or the like is not formed, so that signals (high-frequency signals) of the signal lines can be effectively prevented from leaking from the opening portion **595** to the outside.

[0126] An isolation characteristic improving portion **570** may be disposed between the signal contact member **520**A and the signal contact member **520**B to block at least a part of a region between the signal contact member **520**A and the signal contact member **520**B. In such a manner, since the isolation characteristic improving portion **570** is disposed to block the region between the signal contact member **520**A and the signal contact member **520**B, signals can be more effectively prevented from propagating between the signal contact member **520**A and the signal contact member **520**B. Accordingly, the isolation characteristic can be further improved.

[0127] The isolation characteristic improving portion **570** may extend along an extending direction of the signal contact member **520**A and of the signal contact member **520**B. Accordingly, the region between the signal contact member **520**A and the signal contact member **520**B is effectively blocked by the isolation characteristic improving portion **570**, so that the above-described isolation characteristic can be further improved.

[0128] The isolation characteristic improving portion **570** may be disposed between the signal contact member **520**A and the signal contact member **520**B to block an entirety of the region between the signal contact member **520**A and the signal contact member **520**B. In such a manner, since the region between the signal contact members is completely blocked by the isolation characteristic improving portion **570**, the above-described isolation characteristic can be further improved.

[0129] An area of the isolation characteristic improving portion **570** when viewed in a width direction that is an arrangement direction of the signal contact member **520**A and the signal contact member **520**B may be larger than an area of each of the signal contact member **520**A and the signal contact member **520**B when viewed in the arrangement direction. In such a manner, since the area of the isolation characteristic improving portion **570** is set to be large, signals are more effectively prevented from propagating between the signal contact members by the isolation characteristic improving portion **570**. Accordingly, the isolation characteristic can be further improved.

[0130] The isolation characteristic improving portion **570** may be formed of a member separate from other regions in the second ground connection portion. Accordingly, the degree of freedom in the shape, the raw material, and the disposition of the isolation characteristic improving portion **570** is increased, and signals are more effectively prevented from propagating between the signal contact members by the isolation characteristic improving portion **570** of which the shape, the raw material, and the disposition are appropriately set. Accordingly, the isolation characteristic can be further improved.

[0131] The isolation characteristic improving portion **570** may be connected to a ground bar **581** outer conductors **15** of the coaxial cables **12**A and **12**B. Accordingly, similarly to the center conductor **13** of the coaxial cable **12**A and the center conductor **13** of the coaxial cable **12**B adjacent to each other with the outer conductors **15** sandwiched therebetween, the signal contact member **520**A and the signal contact member **520**B are adjacent to each other with the isolation characteristic improving portion **570** sandwiched therebetween, the isolation characteristic improving portion **570** being electrically connected to the outer conductors **15** through the ground bar **581**, so that similarly to the coaxial cables **12**A and **12**B adjacent to each other, signals can be effectively prevented from propagating between the signal contact member **520**A and the signal contact member **520**B.

REFERENCE SIGNS LIST

[0132] **11**, **511**: coaxial connector device, **12**A: coaxial cable (first coaxial cable), **12**B: coaxial cable (second coaxial cable), **13**: center conductor, **15**: outer conductor, **20**A, **520**A: signal contact member (first signal contact member), **20**B, **520**B: signal contact member (second signal contact member), **30**, **530**: ground contact member, **31**, **531**: annular fitting portion, **34**, **534**: shell portion, **34***b*, **34***c*, **34***d*, **534***b*, **534***c*: clamping portion (barrel portion), **50**, **550**: conductive member, **54**, **570**: isolation characteristic improving portion (second ground connection portion, second connection portion), **55**, **555**: rear ground connection portion (first ground connection portion), **56**, **557**: front ground connection portion (second ground connection portion, first connection portion), **100**, **600**: mating connector device (mating connector), **101**: ground contact portion, **102**, **630**: annular fitting portion (first portion), **103**, **603**: contact portion (second portion), **104**, **604**: signal contact portion (first signal contact portion), **105**, **605**: signal contact portion (second signal contact portion), **581**: ground bar, OP**1**, **595**: opening portion.

1. An electrical connector comprising:
a first signal contact member that is connected to a center conductor of a first coaxial cable, and that is connected to a first signal contact portion of a mating connector mounted on a circuit board;
a second signal contact member that is connected to a center conductor of a second coaxial cable, and that is connected to a second signal contact portion of the mating connector;
a ground contact member to which a ground potential is applied, the ground contact member comprising:
an annular fitting portion that is fitted and connected to a ground contact portion of the mating connector; and
a shell portion that is connected to outer conductors of the first coaxial cable and of the second coaxial cable; and
a conductive member comprising:
a first ground connection portion that is connected to the ground contact member; and
a second ground connection portion that is connected to the ground contact portion of the mating connector,
wherein the second ground connection portion has an isolation characteristic improving portion that extends between the first signal contact member and the second signal contact member in an extension direction thereof and prevents propagation of signals between the first signal contact member and the second signal contact member.

2. The electrical connector according to claim **1**,
wherein the second ground connection portion further includes a front ground connection portion being a flat plate-shaped portion connected to the ground contact portion of the mating connector, and
wherein the isolation characteristic improving portion is configured as a separate member from the front ground connection portion.

3. The electrical connector according to claim **1**,
wherein the conductive member overlaps at least a part of the first signal line, the first signal line comprising the center conductor of the first coaxial cable and the first signal contact member, and at least a part of the second signal line, the second signal line comprising the center conductor of the second coaxial cable and the second signal contact member, when viewed from a direction of fitting to the mating connector.

4. An electrical connector comprising:
a first signal contact member that is connected to a center conductor of a first coaxial cable, and that is connected to a first signal contact portion of a mating connector mounted on a circuit board;
a second signal contact member that is connected to a center conductor of a second coaxial cable, and that is connected to a second signal contact portion of the mating connector;
a ground contact member to which a ground potential is applied, the ground contact member comprising:
an annular fitting portion that is fitted and connected to a ground contact portion of the mating connector; and
a shell portion that to be connected to outer conductors of the first coaxial cable and of the second coaxial cable; and
a conductive member comprising:
a first ground connection portion that is connected to the ground contact member; and
a second ground connection portion that is connected to the ground contact portion of the mating connector,
wherein the second ground connection portion of the conductive member overlaps at least a part of the first signal line, the first signal line comprising the center conductor of the first coaxial cable and the first signal contact member, and at least a part of the second signal line, the second signal line comprising the center conductor of the second coaxial cable and the second signal contact member, when viewed from a direction of fitting to the mating connector.

5. An electrical connector comprising:
a first signal contact member that is connected to a center conductor of a first coaxial cable, and that is connected to a first signal contact portion of a mating connector mounted on a circuit board;
a second signal contact member that is connected to a center conductor of a second coaxial cable, and that is connected to a second signal contact portion of the mating connector;
a ground contact member to which a ground potential is applied, the ground contact member comprising:
a fitting portion that is fitted and connected to a ground contact portion of the mating connector; and
a shell portion that to be connected to outer conductors of the first coaxial cable and of the second coaxial cable; and
a conductive member comprising:
a first ground connection portion that is connected to the ground contact member; and
a second ground connection portion that is connected to the ground contact portion of the mating connector,

wherein the second ground connection portion has an isolation characteristic improving portion that extends between the first signal contact member and the second signal contact member in an extension direction thereof and prevents propagation of signals between the center conductor of the first coaxial cable and the center conductor of the second coaxial cable.

6. An electrical connector comprising:

a first signal contact member that is connected to a center conductor of a first coaxial cable, and that is connected to a first signal contact portion of a mating connector mounted on a circuit board;

a second signal contact member that is connected to a center conductor of a second coaxial cable, and that is connected to a second signal contact portion of the mating connector;

a ground contact member to which a ground potential is applied, the ground contact member comprising:

a fitting portion that is fitted and connected to a ground contact portion of the mating connector; and

a shell portion that to be connected to outer conductors of the first coaxial cable and of the second coaxial cable; and

a conductive member comprising:

a first ground connection portion that is connected to the ground contact member; and

a second ground connection portion that is connected to the ground contact portion of the mating connector,

wherein the second ground connection portion has an isolation characteristic improving portion that extends between the first signal contact member and the second signal contact member in an extension direction thereof and is disposed between the center conductor of the first coaxial cable and the center conductor of the second coaxial cable so as to block a region therebetween.

7. An electrical connector comprising:

a first signal contact member that is connected to a center conductor of a first coaxial cable, and that is connected to a first signal contact portion of a mating connector mounted on a circuit board;

a second signal contact member that is connected to a center conductor of a second coaxial cable, and that is connected to a second signal contact portion of the mating connector;

a ground contact member to which a ground potential is applied, the ground contact member comprising:

a fitting portion that is fitted and connected to a ground contact portion of the mating connector; and

a shell portion that to be connected to outer conductors of the first coaxial cable and of the second coaxial cable; and

a conductive member comprising:

a first ground connection portion that is connected to the ground contact member; and

a second ground connection portion that is connected to the ground contact portion of the mating connector,

wherein the second ground connection portion overlaps at least a part of the first signal line, the first signal line comprising the center conductor of the first coaxial cable and the first signal contact member, and at least a part of the second signal line, the second signal line comprising the center conductor of the second coaxial cable and the second signal contact member, when viewed from a direction intersecting with an extension direction of the first coaxial cable and the second coaxial cable and an arrangement direction of the first signal contact member and the second signal contact member.

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